

Deep Learning as Generalization Machinery

From Task-Specific Mappings to Reusable Representations and Predictive Relational Structure

Zixiong Wang
Shaoxing University

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Contact: wangzixiong@usx.edu.cn; teddywang0405@gmail.com

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Abstract

Deep learning is usually described through architectures, benchmarks, datasets, and scale: convolutional networks, Transformers, ImageNet, BERT, GPT, larger models, larger corpora, and larger compute. This paper argues that such descriptions miss a deeper organizing principle. The major transitions in deep learning are transitions in what models learn to generalize.

This paper proposes a three-level framework of generalization in deep learning. Level 1 is supervised cross-sample generalization: models learn task-specific mappings from labeled examples and apply them to unseen samples. This regime made deep learning practically powerful, with supervised CNNs in computer vision and early neural NLP systems demonstrating that learned neural representations could replace hand-engineered pipelines in high-dimensional perception and language tasks.

Level 2 is self-supervised representation generalization: models learn reusable internal representations from data-derived supervision and transfer them across downstream tasks. BERT made this regime decisive in NLP by turning self-supervised pretraining into a general representation substrate. Self-supervised vision and multimodal models such as SimCLR, DINO, MAE, and CLIP extended the same logic beyond language, showing that representations themselves could become reusable infrastructure.

Level 3 is autoregressive and predictive high-dimensional relational generalization: models learn predictive structure over broad symbolic or multimodal streams, and the predictive process itself becomes a capability interface. GPT-style models made this transition decisive by turning next-token prediction over language and code into a general interface for prompting, in-context learning, instruction following, reasoning traces, code generation, dialogue, and tool use.

The distinction between BERT and GPT is therefore not merely architectural. BERT makes representations reusable; GPT makes prediction interactive. This difference marks a transition from representation transfer to predictive interaction. The proposed framework explains deep learning progress as an expansion of the object of generalization: from task mappings, to reusable representations, to predictive relational structures. It also suggests that future advances will depend not only on scale, but on discovering richer generalization objects and more powerful interfaces through which learned structure can be used.

1. Introduction

Deep learning is fundamentally a project of generalization.

This statement means more than the ordinary claim that a trained model should perform well on held-out test samples. Test-set generalization is important, but it is only the narrowest expression of a deeper principle. In its most consequential forms, deep learning is a method for extracting structure from finite experience and making that structure effective beyond the exact conditions under which it was observed. A deep model is not merely a function approximator that maps inputs to labels; it is

also a mechanism for compressing, reorganizing, and transferring the latent regularities of data into broader spaces of use.

The central question of deep learning is therefore not only whether a model can fit a dataset, or whether it can interpolate among unseen samples drawn from a similar distribution. The deeper question is what level of structure the model can generalize, and how far that structure can travel beyond its original training setting. Once this question is placed at the center, the major transitions in deep learning no longer look like ordinary improvements in task performance. They look like transitions in the level of generalization itself.

This paper develops a position around that idea. It argues that the major progress of deep learning can be understood as a sequence of upward transitions in generalization. Deep learning first became practically powerful by learning task-specific mappings that generalized from training samples to unseen samples. It then became foundational by learning reusable representations from data itself and transferring those representations across many downstream tasks. In its most consequential current form, it has begun to learn high-dimensional relational structure through autoregressive or predictive self-supervision, where the model is not only recognizing patterns already present in the input but also learning to predict future tokens, future states, future representations, or future continuations of structured data [1]–[5].

The word **generalization** is therefore used in this paper in an expanded sense. In conventional machine learning, generalization usually refers to the relationship between training performance and test performance. This meaning remains essential, and modern deep networks have made it even more difficult to understand: sufficiently large neural networks can fit random labels, while still generalizing remarkably well on real data [6]. However, the present paper uses generalization in a broader and more historical sense. Generalization is the capacity of a learning system to infer transferable structure from finite observations and to make that structure effective beyond the original boundary of supervision, task, modality, or observed sequence.

This broader meaning is necessary because the most important transitions in deep learning cannot be fully captured by test accuracy alone. A supervised ImageNet classifier, a BERT-style language representation model, and a GPT-style autoregressive model all generalize, but they do not generalize the same kind of object. The first generalizes task-specific decision boundaries from seen samples to unseen samples. The second generalizes reusable representations from self-supervised pretraining to multiple downstream tasks. The third, in its most successful current instantiation, generalizes predictive structure over language, code, symbols, interaction traces, and potentially other temporally or relationally organized data into open-ended task behavior and strong human-like reasoning capability.

This paper proposes three levels of generalization in deep learning.

The first level is **supervised cross-sample generalization**. In this regime, the model is trained for a single human-defined task, usually with labeled data, and its central achievement is to perform that same task on previously unseen samples. This is the level at which deep learning first became practically undeniable. The historical success of large supervised vision models belongs to this level [1], [7].

The second level is **self-supervised representation generalization**. In this regime, the model is no longer trained only as a task-specific predictor. It learns reusable representations from the structure of data itself through masking, contrast, reconstruction, alignment, or other self-supervised objectives, and those representations can be adapted to multiple downstream tasks. This level marks the rise of foundation-like representation models. BERT is the canonical example in natural language processing; SimCLR, DINO, and MAE represent corresponding developments in computer vision; CLIP shows that the same level can also extend across modalities through representation alignment [8].

The third level is **autoregressive or predictive high-dimensional relational generalization**. This level should not be reduced to the claim that large language models generate open-ended text. The broader definition is that, at Level 3, models learn complex high-dimensional relations by predicting future tokens, future states, future representations, or future world dynamics. The successful case we currently understand best is the large language model, because language, code, mathematical notation, dialogue, documentation, and symbolic text are not ordinary data streams. They are externalized traces of human cognition, problem solving, explanation, planning, communication, and tool use. When an autoregressive model is trained at sufficient scale on such traces, next-token prediction becomes more than surface continuation; it becomes a pressure to model the relational structure behind human-produced symbolic activity [3]–[5], [9].

For this reason, LLMs should not be described merely as models that have learned open-ended human-like behavior. That description is partially correct, but it is not strong enough. State-of-the-art LLMs have also demonstrated strong human-like reasoning ability in many domains, including mathematics, coding, symbolic reasoning, scientific problem solving, and multi-step planning, especially when reasoning is elicited or strengthened through chain-of-thought prompting, instruction tuning, reinforcement learning, and test-time computation [5], [10]–[12]. They are not complete human intelligence, and they should not be described as such. At the same time, reducing them to behavioral imitation would underestimate what the evidence now shows. A more accurate claim is that LLMs have learned a powerful subset of human-intelligence-like structure externalized in language, code, symbols, and textual interaction, and that this structure can be released as both open-ended task behavior and strong reasoning capability.

LLMs, however, are not the whole definition of Level 3. They are the first successful large-scale instantiation of it. The deeper regime is predictive high-dimensional relational generalization. A future world model might learn physical dynamics by predicting future latent states. A video model might learn temporal and causal regularities by predicting future frames or representations. A multimodal agent might learn the structure of perception and action by predicting the consequences of interaction. A JEPA-like system might learn semantic world structure by predicting missing or future representations rather than reconstructing pixels [13]–[17]. These systems would not merely extend LLMs; they would instantiate the same deeper principle in other domains, namely the use of predictive self-supervision to learn relations too complex to be captured by ordinary supervised task labels.

This framework treats deep learning progress not as an accumulation of isolated techniques, and not merely as a consequence of scale, but as an expansion in what neural networks are able to generalize. Scale matters profoundly, but scale is not neutral. It amplifies the kind of generalization made accessible by the training objective, the data distribution, the architecture, and the output interface. Scaling a supervised classifier strengthens one kind of generalization. Scaling a masked representation model strengthens another. Scaling an autoregressive predictive model over human symbolic traces opens a different space of capabilities altogether.

This distinction is especially important for understanding the historical relation between BERT and GPT. BERT and GPT-1 appeared in nearly the same period of large-scale self-supervised NLP pretraining, but they unfolded into fundamentally different generalization regimes. BERT crystallized the second level: self-supervised language representation generalization. GPT-1 planted the seed of the third: autoregressive predictive generalization over symbolic sequences, which was later expanded through GPT-2, GPT-3, instruction tuning, ChatGPT, code models, chain-of-thought reasoning, and reinforcement-learning-based reasoning systems. Their difference is therefore not merely encoder versus decoder, and it is not merely understanding versus generation. It is a difference between representation-level generalization and predictive high-dimensional relational generalization.

The present paper is a conceptual position paper. It does not introduce a new model, benchmark, or theorem. Its purpose is to articulate a general framework for understanding why deep learning has advanced through discontinuous changes in capability rather than only through smooth improvements in task performance. Existing work has already shown that deep learning forces us to rethink generalization [6], that intelligence should be evaluated in relation to generalization difficulty and scope rather than only task skill [18], that foundation models trained on broad data at scale can be adapted to many downstream tasks [8], and that prompting constitutes a new paradigm for using pretrained language models [19]. This paper builds on those materials, but proposes a distinct synthesis: deep learning should be understood as a progression from sample-level generalization, to representation-level generalization, to predictive high-dimensional relational generalization.

The thesis can be stated compactly:

Deep learning is not merely a class of scalable function approximators. It is a machinery of generalization. Its major historical transitions are transitions in what can be generalized: from samples, to representations, to predictive high-dimensional relations.

1.1 Contributions

This paper makes three main contributions.

First, it proposes a three-level framework for understanding deep learning progress as a progression in the object of generalization. The framework distinguishes supervised cross-sample generalization, self-supervised representation generalization, and autoregressive or predictive high-dimensional relational generalization.

Second, it uses this framework to reinterpret several major historical transitions in deep learning. AlexNet and early neural NLP are treated as decisive Level 1 milestones; BERT-style pretraining is treated as the crystallization of Level 2 in NLP; and GPT-style autoregressive modeling is treated as the first decisive large-scale realization of Level 3.

Third, it clarifies why BERT and GPT should not be understood merely as different Transformer architectures. Their deeper difference lies in how learned structure becomes usable: BERT makes representations reusable, while GPT makes prediction interactive. This distinction also helps explain why future progress may depend not only on larger models, but on richer generalization objects and stronger capability interfaces.

2. A Three-Level Framework of Generalization

The previous section introduced the central claim of this paper: deep learning should be understood as a machinery of generalization, and its major historical transitions should be understood as transitions in what can be generalized. This section makes that claim more precise by defining a **generalization regime** and then specifying three levels of generalization that have shaped the development of deep learning.

A **generalization regime** is not the same thing as a model architecture. It is the broader learning configuration that determines what kind of structure a model is trained to extract, how that structure is stored or expressed, and how it can be used beyond the original training setting. A regime includes at least four components: the training signal, the type of data, the form of internal structure learned by the model, and the interface through which the learned structure is later used. Two models may share a similar architecture but belong to different generalization regimes if their training objectives and capability interfaces lead them to generalize different kinds of structure.

This distinction is necessary because deep learning progress cannot be explained only by saying that models became larger, deeper, or more computationally expensive. Scale matters, but scale acts on a regime. It expands the capabilities that the training objective and data distribution make available. A supervised classifier and an autoregressive language model may both benefit from scale, but they do not benefit from it in the same way, because they are trained to extract and deploy different kinds of structure. A generalization regime is therefore a more fundamental unit of analysis than a parameter count or a single architectural choice.

The three levels proposed in this paper can be understood as increasingly broad answers to a single question: **what is being generalized?** At the first level, the model generalizes a task-specific mapping from seen samples to unseen samples. At the second level, the model generalizes reusable representations from self-supervised pretraining to multiple downstream tasks. At the third level, the model generalizes predictive high-dimensional relational structure from sequences, states, representations, or other structured data into future continuations, open-ended task behavior, and, in the case of LLMs, strong human-like reasoning capabilities.

2.1 Level 1: Supervised Cross-Sample Generalization

The first level is supervised cross-sample generalization. In this regime, the model is trained for a specific task, usually with human-provided labels, and its main achievement is to perform the same task on unseen samples. The model learns a mapping from inputs to outputs within a fixed task definition, and the central measure of success is whether the learned mapping transfers from the training set to the test set.

This is the level most closely associated with the conventional meaning of generalization in machine learning. A model is trained on a finite set of labeled examples, and it is expected to perform correctly on new examples drawn from a similar distribution. This kind of generalization is far from trivial. Modern deep networks have made it theoretically difficult to explain, because they can be expressive enough to memorize arbitrary labels while still generalizing well on real data [6]. Nevertheless, the generalization object at this level remains relatively narrow: the model generalizes across samples within a single task.

The historical importance of Level 1 should not be underestimated. This is the level at which deep learning first became practically decisive in domains such as computer vision and speech recognition [1], [20]. It demonstrated that deep neural networks could learn useful structure directly from high-dimensional data and could outperform hand-engineered systems on difficult real-world tasks. But Level 1 remains task-bound. A model trained in this regime may learn powerful features, and those features may later be reused, but the regime itself is organized around a single task and a single performance objective.

In the language of this paper, Level 1 made deep learning useful. It established that neural networks could generalize within difficult supervised tasks, but it did not yet establish that one model could serve as a reusable basis for many tasks, or that predictive learning over rich structured data could produce open-ended behavior and strong reasoning.

2.2 Level 2: Self-Supervised Representation Generalization

The second level is self-supervised representation generalization. In this regime, the model is no longer trained only as a task-specific predictor. Instead, it learns representations from the structure of data itself, usually through objectives such as masking, contrast, reconstruction, alignment, or other forms of self-supervision [2], [21]–[24]. The learned representation can then be adapted to multiple downstream tasks.

The key transition from Level 1 to Level 2 is the movement from task-specific mapping to reusable structure. A Level 2 model does not merely learn how to solve one labeled task. It learns an internal representation that can support many tasks, including tasks that were not explicitly present during pretraining. For this reason, self-supervised learning became central to the rise of foundation-like models. The Stanford foundation model report defines foundation models as models trained on broad data at scale, generally using self-supervision, that can be adapted to a wide range of downstream tasks [8]. This definition captures the essence of Level 2: the model learns from broad data and becomes useful beyond a single supervised objective.

Level 2 changes the meaning of generalization. The model still generalizes across samples, but that is no longer the central achievement. Its deeper achievement is that the representation learned from one large pretraining process can be transferred across tasks. In natural language processing, this level is exemplified by BERT-style language representation learning. In computer vision, it is exemplified by self-supervised visual representation learning methods such as contrastive learning, self-distillation, and masked autoencoding. In multimodal learning, it appears in models that align representations across modalities, such as image-text representation models.

This level also explains why the concept of a foundation model became necessary. Once a model is trained on broad data and adapted to many downstream tasks, it is no longer merely a task model. It becomes an infrastructure-like representation system. Its value lies not only in one benchmark score, but in the fact that many task-specific systems can be built on top of it. This is a major expansion in the object of generalization: from unseen samples within one task to reusable representations across many tasks.

However, Level 2 remains primarily representation-centered in its capability interface. The model usually requires fine-tuning, a task head, a linear probe, retrieval machinery, reranking, or some downstream pipeline to convert its representation into task behavior. This is not a weakness; it is the defining form of its power. Level 2 made deep learning foundational because it transformed models into reusable representation bases. But it did not yet fully establish the open-ended predictive and generative interface that characterizes Level 3.

2.3 Level 3: Autoregressive or Predictive High-Dimensional Relational Generalization

The third level is autoregressive or predictive high-dimensional relational generalization. This level is broader than large language modeling, although large language models are its first successful large-scale instantiation. At this level, the model learns complex relational structure by predicting future tokens, future states, future representations, or future dynamics. The central idea is that prediction over rich structured data can force a model to infer deeper relations than ordinary supervised labels or static representations can capture.

This definition deliberately keeps Level 3 open. LLMs are not the whole category. They are the first case in which this regime has been fully demonstrated at civilizational scale. Language and code are unusually powerful substrates because they contain externalized traces of human knowledge, cognition, reasoning, planning, communication, and problem solving. When an autoregressive model is trained to predict future tokens across sufficiently broad language and code distributions, it can learn not only surface linguistic continuation but also the deeper relational structure behind human symbolic activity. As a result, LLMs can exhibit both open-ended task behavior and strong human-like reasoning ability in many domains, especially after instruction tuning, chain-of-thought prompting, reinforcement learning, and test-time computation [5], [10]–[12].

The concept of Level 3 should nevertheless extend beyond LLMs. A world model that predicts future latent states, a video model that predicts future visual representations, or a multimodal agent that

predicts the consequences of interaction may all instantiate the same deeper regime if they learn high-dimensional relational structure through predictive self-supervision. In these cases, the predictive target may not be text. It may be a future state, a future action-conditioned transition, a future representation, or a future configuration of the environment. What unifies these systems is not the modality, but the use of prediction as a pressure to learn relations that are too complex to be reduced to ordinary supervised task labels.

Level 3 therefore marks a deeper shift in the meaning of generalization. The model is no longer only expected to perform a fixed supervised task, and it is no longer only expected to provide reusable representations for downstream adaptation. It begins to learn structured continuations of the world, or of symbolic traces of the world, and to make those continuations usable through a generative or predictive interface. In the case of LLMs, this interface appears as dialogue, code generation, mathematical reasoning, planning, and instruction following. In future systems, it may appear as world-state prediction, embodied planning, video understanding, or multimodal interaction.

The importance of Level 3 is that it reveals a broader possibility for deep learning. If Level 1 shows that neural networks can generalize across samples, and Level 2 shows that they can generalize reusable representations across tasks, then Level 3 shows that predictive self-supervision can generalize high-dimensional relations that organize behavior, reasoning, and possibly world dynamics. This does not imply that current LLMs are complete human intelligence, nor does it imply that future world models are already solved. It means that deep learning has entered a regime in which prediction itself can become a route toward learning highly structured relations.

Table 1 summarizes the three proposed levels.

Table 1. Three Levels of Generalization in Deep Learning

Level 1

- **Generalization Regime:** Supervised cross-sample generalization
- **Primary Learning Signal:** Human-provided labels or task-specific supervision
- **Generalization Object:** A task-specific mapping from seen samples to unseen samples
- **Typical Capability Interface:** Fixed task output, such as classification or regression
- **Representative Examples:** Large supervised vision and speech models
- **Historical Meaning:** Deep learning becomes practically useful

Level 2

- **Generalization Regime:** Self-supervised representation generalization
- **Primary Learning Signal:** Data-derived objectives such as masking, contrast, reconstruction, or alignment
- **Generalization Object:** Reusable representations that transfer across downstream tasks
- **Typical Capability Interface:** Fine-tuning, task heads, linear probes, retrieval, reranking, downstream pipelines
- **Representative Examples:** BERT-style language representation models; self-supervised vision models; multimodal alignment models such as CLIP
- **Historical Meaning:** Deep learning becomes foundational

Level 3

- **Generalization Regime:** Autoregressive or predictive high-dimensional relational generalization
- **Primary Learning Signal:** Prediction of future tokens, states, representations, or dynamics
- **Generalization Object:** High-dimensional relational structure over sequences, symbols, states, or world dynamics

- **Typical Capability Interface:** Generative or predictive interfaces; open-ended task behavior; strong reasoning in the LLM case; future world-state prediction in broader cases
- **Representative Examples:** GPT-style LLMs; reasoning models; future world models, video models, JEPA-like predictive systems
- **Historical Meaning:** Deep learning begins to generalize predictive structure that can support open-ended behavior, reasoning, and potentially world modeling

Table 1 summarizes the conceptual framework proposed in this paper. The three levels should not be read as a simple chronological taxonomy of model architectures. They are regimes of generalization. A later model can contain components of earlier levels, and a single system may combine multiple regimes. The table identifies the dominant form of generalization that made each level historically significant.

This framework also clarifies the role of scale. Scale alone does not define a level. Instead, scale amplifies the capability space made available by a generalization regime. When scale is applied to a supervised task model, it can produce stronger task performance. When scale is applied to a self-supervised representation model, it can produce more transferable representations. When scale is applied to an autoregressive or predictive model trained on rich relational data, it can produce a qualitatively different range of capabilities, including open-ended generation and, in the LLM case, strong human-like reasoning. The key variable is therefore not scale in isolation, but scale operating inside a particular generalization regime.

The rest of the paper develops each level in detail and then focuses on the central historical case: BERT and GPT. BERT represents one of the clearest successes of Level 2 because it crystallized self-supervised language representation learning. GPT represents the first successful path into Level 3 because its autoregressive predictive objective, when scaled and later shaped through post-training, opened a different space of capability. The distinction between these two routes is one of the strongest pieces of evidence that deep learning progress is not merely a story of larger models, but a story of higher levels of generalization.

3. Level 1: Supervised Cross-Sample Generalization in Vision and Early Neural NLP

3.1 The First Form of Deep Learning Generalization

The first level of generalization is supervised cross-sample generalization. In this regime, a model is trained for a human-defined task, usually with labeled data, and its central achievement is to perform the same task on samples that were not present in the training set. The model is not yet expected to become a reusable representation base for many downstream tasks, and it is not yet expected to generate open-ended behavior, reason through symbolic problems, or predict the future structure of the world. Its task is narrower, but historically decisive: it must learn a task-specific structure from finite examples and make that structure work on unfamiliar cases.

This is the form of generalization closest to classical supervised learning, but it is not shallow. In deep learning, even cross-sample generalization is theoretically difficult. Large neural networks can have enough capacity to fit arbitrary labels, yet they can still generalize well on real data [6]. Level 1 is therefore already a serious generalization regime. It involves high-dimensional inputs, large parameter spaces, optimization dynamics, inductive biases, learned features, and dataset structure interacting in ways that cannot be reduced to a simple picture of fitting a fixed decision boundary.

The object of generalization at this level is a task-specific mapping. In image classification, the model generalizes from labeled training images to unseen images. In speech recognition, it generalizes from

labeled acoustic examples to new utterances. In early neural NLP, it generalizes from labeled or parallel text examples to unseen sentences. The task remains fixed, while the samples vary. The unknown space is primarily a space of new instances rather than a space of new tasks, new modalities, or open-ended behaviors.

Computer vision provides one of the clearest historical demonstrations of this first level because the supervised CNN line made the scale problem unusually visible. The decisive issue was not simply whether convolutional neural networks could work. They could. The deeper issue was whether deep neural networks could be scaled into a regime where they could dominate large-scale natural image recognition, a task that had previously been ruled by hand-designed features, classical machine learning pipelines, and carefully engineered visual representations.

3.2 AlexNet and the Public Breaking of the Scale Barrier

AlexNet is the decisive event in the computer-vision story of Level 1. Its importance is often summarized by saying that it won ImageNet in 2012, but that description is too small. AlexNet was not merely a high-performing image classifier. It was the moment when a long-standing scale barrier in deep learning was broken publicly, decisively, and in front of the entire computer vision community.

This history has to be stated carefully. AlexNet did not invent convolutional neural networks. The CNN lineage had already been built through earlier work such as LeNet and gradient-based document recognition, where many of the essential principles of later CNNs were already present: local receptive fields, convolution, weight sharing, subsampling, hierarchical feature extraction, and end-to-end supervised learning [25]. AlexNet also did not invent GPU neural network training. Before 2012, researchers had already explored GPU acceleration for neural networks and CNNs in document processing, large-scale unsupervised deep learning, and image classification [26]–[30]. But these qualifications do not reduce the historical significance of AlexNet. They clarify it. AlexNet was the first work to bring together a very large supervised CNN, ImageNet-scale natural image data, GPU computation, and a modern training recipe into a result large enough to overturn the dominant visual recognition paradigm.

The central background was the scale bottleneck. Before AlexNet, practical supervised CNN research was not absent, but it operated in a very different scale regime. LeNet-5 was a foundational CNN system for handwritten digit and document recognition, but it had roughly sixty thousand trainable parameters and worked on small character images rather than million-image natural image classification [25]. Later multi-stage ConvNet research from the LeCun group explored object recognition architectures, rectification, local contrast normalization, pooling, and learned feature hierarchies, but these systems were still studied in object recognition settings far smaller than ImageNet-scale 1000-class natural image classification [31]. Sermanet and LeCun applied multi-scale convolutional networks to traffic sign recognition and explicitly contrasted ConvNets, which learn feature hierarchies from data, with hand-crafted features such as HOG and SIFT; but the input images were small traffic-sign crops and the problem was much narrower than ImageNet [32]. Cireřan and collaborators built fast GPU CNNs that achieved strong or state-of-the-art results on MNIST, NORB, CIFAR-10, and traffic sign benchmarks, and their work was an important pre-AlexNet demonstration that supervised CNNs and GPUs could be powerful [28], [29]. Yet these works still did not establish what AlexNet established: that a huge supervised CNN could be trained end-to-end on high-resolution ImageNet-scale natural images and defeat the strongest hand-engineered systems by a very large margin.

This is the point that must not be lost. Before AlexNet, the supervised CNN field had important successes, but the mainstream practical evidence for CNNs was largely concentrated in smaller or more constrained regimes: handwriting, document recognition, small-object benchmarks, traffic signs,

CIFAR-like images, and other settings where input resolution, dataset size, category structure, or benchmark visibility did not yet match ImageNet. The models themselves were correspondingly much smaller or operated in much narrower task settings than AlexNet. The field had strong reasons to believe in CNNs, but it did not yet have a public, large-scale, natural-image, thousand-class result that forced the entire vision community to reorganize its expectations.

The compute bottleneck was the material reason for this limitation. Large natural image datasets, high-resolution inputs, large convolutional networks, and long supervised training runs demanded far more computation and memory than the older CPU-centered workflow could comfortably support. The AlexNet paper states this directly: although CNNs had attractive qualities, applying them at large scale to high-resolution images had been prohibitively expensive, and modern GPUs together with optimized 2D convolution made it possible to train “interestingly-large” CNNs [1]. The paper also states that the size of the network was limited mainly by available GPU memory and tolerable training time, and that their network took five to six days to train on two NVIDIA GTX 580 3GB GPUs [1]. This was not a minor engineering footnote. It was the practical condition that allowed the decisive scale experiment to exist.

The GPU story before AlexNet strengthens this point rather than weakening it. Raina, Madhavan, and Ng had shown in 2009 that GPUs could make very large neural networks computationally feasible by reducing the training time of a four-layer deep belief network with 100 million free parameters from several weeks to about a day [27]. Cireşan and collaborators showed that GPU implementations made deep CNN training dramatically faster and allowed strong results on several visual benchmarks [28], [29]. In a later NVIDIA interview, Cireşan described DNNs as massively parallel structures, stated that training on normal CPU clusters was difficult because of communication costs, and said that stochastic gradient descent could run up to 40 times faster on a GPU than on a CPU; he also noted that before receiving a GTX 280, he had to use small networks even while running experiments on 20 computers [30]. These sources show that GPU computation was already understood as a powerful route out of the CPU bottleneck. What AlexNet did was to use that route at the right scale, on the right supervised benchmark, with the right model and training system.

The conceptual bottleneck followed from the compute bottleneck. If the community cannot easily train much larger networks on much larger datasets, then it cannot cleanly answer the central question: are deep networks underperforming because neural networks are intrinsically weak for large-scale vision, because the architecture and training tricks at the available scale are not good enough, or because the models have not yet been scaled into the regime where their strengths become visible? This uncertainty was not simply conservatism. It was a rational uncertainty created by a technical ceiling. When the available experiments are mostly small compared with the real target problem, it is hard to know whether one should search for better small-scale tricks or break the scale ceiling itself.

AlexNet broke that uncertainty by running the scale experiment that the field had not yet seen at ImageNet level. The model was trained on approximately 1.2 million high-resolution ImageNet training images for a 1000-class object classification task, with 50,000 validation images and 150,000 test images [1], [33]. It had about 60 million parameters and 650,000 neurons, consisting of five convolutional layers, three fully connected layers, and a final 1000-way softmax [1]. It was trained on two NVIDIA GTX 580 GPUs, with the network split across them because of GPU memory limits, and training took approximately five to six days [1]. Relative to LeNet-scale practical CNNs, this was not a routine enlargement. It was a jump from tens of thousands of parameters in narrow document recognition to tens of millions of parameters in large-scale natural image classification. Relative to pre-AlexNet GPU CNNs on smaller benchmarks, it moved supervised CNNs into a far more demanding regime: larger data, richer images, more categories, higher benchmark visibility, and stronger classical baselines.

The model also mattered because it turned scale into trainable scale. AlexNet used ReLU nonlinearities, which allowed faster training than saturating nonlinearities and made experimentation with large networks more practical [1]. It used random crops, horizontal reflections, and RGB color perturbations to augment the data and reduce overfitting [1]. It used dropout in the fully connected layers, local response normalization, overlapping pooling, stochastic gradient descent with momentum, and weight decay [1]. These choices were not decorative. They were the recipe that made a 60-million-parameter supervised CNN trainable, regularized, and effective on ImageNet. AlexNet was not brute-force scaling; it was a full training system built to make aggressive scaling work.

The result was a field-level rupture. On ILSVRC-2010, AlexNet achieved 37.5% top-1 test error and 17.0% top-5 test error. The best previously published results were 45.7% top-1 and 25.7% top-5 error from a system based on Fisher Vectors computed from densely sampled features; the best ILSVRC-2010 competition result had been 47.1% top-1 and 28.2% top-5 error from a sparse-coding approach [1]. In ILSVRC-2012, AlexNet's ensemble achieved 15.3% top-5 test error, while the second-best entry achieved 26.2% using an approach based on averaging classifiers trained on Fisher Vectors from different dense features [1]. In accuracy terms, AlexNet reached 84.7% top-5 accuracy, while the runner-up reached 73.8%. The top-5 error gap was 10.9 percentage points. This was not an incremental improvement inside an existing paradigm. It was a visible break in the empirical order of large-scale visual recognition.

The nature of the defeated systems is essential. AlexNet did not merely add a better classifier to the dominant feature-engineering pipeline. It displaced that pipeline. Before AlexNet, strong visual recognition systems relied heavily on human-designed descriptors, dense feature extraction, sparse coding, Fisher Vectors, and classifier ensembles. AlexNet instead trained a large supervised neural network end-to-end on raw image data, allowing the visual hierarchy to be learned inside the model. That is why the result had such force. It did not simply improve ImageNet; it changed the default answer to where visual representations should come from.

In this sense, AlexNet should be understood as a generalization event. It showed that a large supervised neural network could learn high-dimensional perceptual structure directly from data and generalize that structure across unseen natural images better than the best feature-engineered systems. It also showed that scale was not secondary to deep learning. Scale became a first-order scientific variable. When data scale, model scale, compute scale, and training technique were aligned, the system crossed into a stronger supervised generalization regime.

The deeper lesson of AlexNet was not that larger models automatically solve every problem. AlexNet was not scale alone. It was scale under the right inductive bias, the right benchmark, the right data regime, the right compute substrate, and the right optimization recipe. But it established a principle that later deep learning would repeatedly rediscover: if the model class is appropriate and the training pipeline is strong enough, scale can reveal capabilities that smaller experiments cannot expose. In Level 1, this capability was large-scale supervised cross-sample generalization in vision. In Level 2, the same logic would reappear as large-scale self-supervised representation learning. In Level 3, it would reappear again as autoregressive and predictive high-dimensional relational generalization.

For this reason, AlexNet helped restart the modern deep learning wave. LeCun, Bengio, and Hinton later described the 2012 ImageNet breakthrough as a result that almost halved the error rate for object recognition and precipitated the rapid adoption of deep learning by the computer vision community [7]. That statement captures the historical magnitude of the event. AlexNet did not merely win a competition. It changed what researchers believed was worth attempting. It made it much harder to dismiss deep learning as a fragile or niche technique, and much easier to believe that the underperformance of earlier neural systems had partly reflected the fact that the field had not yet been able to train sufficiently large models on sufficiently large data with sufficiently strong computation.

In the framework of this paper, AlexNet is therefore a Level 1 milestone because its dominant achievement was still supervised cross-sample generalization. It did not become a foundation model. It did not become a multimodal assistant. It did not perform open-ended reasoning. It learned a task-specific mapping from images to object categories. But the scale, difficulty, and visibility of that task made the result historically decisive. AlexNet proved that deep neural networks could become dominant supervised generalizers in high-dimensional perception, and it made scale central to the research logic of modern deep learning.

3.3 The Maturation of Supervised CNN Scaling After AlexNet

After AlexNet, the supervised CNN line became a research program for scaling Level 1 more effectively. Later models did not simply repeat AlexNet; they explored different dimensions of supervised scale. Network in Network increased the expressive power of local computation through multilayer perceptron convolutional layers and introduced global average pooling as a lighter and more structurally meaningful classification head [34]. VGG made depth a clean design axis by stacking small 3×3 filters into 16- and 19-layer networks [35]. GoogLeNet used the Inception architecture to increase depth and width under a controlled computational budget, winning the ILSVRC 2014 classification challenge [36]. ResNet addressed the degradation problem of very deep networks through residual learning and won the ILSVRC 2015 classification task with an ensemble top-5 error of 3.57% [37].

These models matter because they show that AlexNet was not a one-off accident. Once AlexNet made large supervised CNNs credible, the field learned to scale the same first-level regime along multiple axes: local expressivity, depth, computational efficiency, and optimization stability. The dominant objective remained supervised cross-sample generalization. The models became deeper, cleaner, more modular, and easier to optimize, but they were still primarily trained to map labeled images to object categories or related supervised outputs.

This boundary is important. Level 1 made deep learning practically dominant in perception, but it did not yet make deep learning foundational in the later sense, and it did not yet create predictive systems capable of open-ended behavior or strong reasoning. It established the first great form of deep learning generalization: large neural networks could learn task-specific structures from high-dimensional data and generalize them to unseen samples. The next section will show that this first level was not only a computer vision story. Early neural NLP and supervised sequence learning were also central expressions of supervised cross-sample generalization.

3.4 Early Neural NLP and Supervised Sequence Generalization

The computer-vision story explains the most visible breakthrough of Level 1, but it does not exhaust the first level of deep learning generalization. Natural language processing was the other major domain in which neural networks began to replace feature-engineered systems with learned task-specific representations. The historical pattern was different from computer vision. NLP did not have a single AlexNet-like event that publicly broke the field in one benchmark. Instead, the transition happened through a sequence of advances that gradually made neural representations, end-to-end learning, and sequence modeling central to supervised language tasks.

The core point is still the same. Early neural NLP belonged to Level 1 because its dominant regime remained task-bound. A model was trained to classify sentiment, assign part-of-speech tags, identify named entities, predict semantic roles, parse syntax, or translate a source sentence into a target sentence. The model could learn useful internal representations, and many of those representations later became important for Level 2 and Level 3. But the original task setting remained supervised or conditionally supervised cross-sample generalization: learn from labeled examples, annotated sequences, or parallel sentence pairs, then perform the same task on unseen examples.

This matters because NLP before deep learning was deeply shaped by sparse symbolic features and task-specific engineering. Traditional systems often depended on n-gram statistics, lexicons, linguistic preprocessing, hand-designed features, phrase tables, alignment models, and separate pipelines for different tasks. These systems could be strong, but their generalization was constrained by the sparsity of language and by the amount of structure humans had to design outside the model. Neural NLP began to change this arrangement by making language structure increasingly learnable inside the model.

One of the earliest foundations was the neural probabilistic language model. Bengio, Ducharme, Vincent, and Jauvin proposed learning distributed word representations together with a probability function over word sequences [38]. The significance of this work was not simply that it improved language modeling. Its deeper importance was that it gave neural NLP a way to fight the curse of dimensionality in language. If words are represented as points in a continuous space, then a model can share statistical strength across semantically or syntactically related words and contexts. A sentence no longer teaches the model only about itself; it also informs nearby regions of representation space. This was not yet Level 2 in the later foundation-model sense, but it was already a crucial step away from purely sparse symbolic generalization.

Collobert et al. then showed that neural networks could be used across several classical NLP tasks with much less dependence on task-specific feature engineering [39]. Their work on *Natural Language Processing (Almost) from Scratch* proposed a relatively unified neural architecture for tasks such as part-of-speech tagging, chunking, named entity recognition, and semantic role labeling [39]. The historical meaning of this line is that neural NLP began to internalize feature learning. Instead of requiring a different hand-engineered feature pipeline for every task, a neural model could learn useful linguistic representations and apply them to multiple supervised language problems. This was still Level 1 because the tasks remained fixed, but it was a richer form of Level 1 than a single isolated classifier.

The same pattern appeared in sentence-level and compositional tasks. Neural models for sentiment classification, sentence classification, and related tasks showed that learned word representations and neural architectures could generalize across unseen sentences without the same degree of manual feature design [40]. These models did not yet create a general language foundation model. They were still trained and evaluated on specific supervised tasks. But they strengthened the first-level regime by showing that task-specific language systems could learn their own internal structure rather than relying entirely on handcrafted features.

The strongest expression of Level 1 in NLP was neural machine translation. Machine translation had long been dominated by statistical machine translation, especially phrase-based systems built from phrase tables, alignment models, language models, reordering components, and log-linear feature combinations [41]. These systems were powerful, but they were also pipeline systems. Much of the structure of translation was distributed across engineered components outside a single neural model.

Neural machine translation changed the form of the problem. Sutskever, Vinyals, and Le showed that a multilayer LSTM sequence-to-sequence model could map an input sequence to an output sequence and achieve strong results on WMT-14 English-to-French translation [42]. The model learned from parallel sentence pairs and generalized to unseen sentence pairs. It belongs to Level 1 because the task remained translation, the supervision came from paired examples, and the model's success was measured by performance on new examples of the same task. But it was a very important form of Level 1 because the task was structured, sequential, and much richer than ordinary classification.

Attention then made neural sequence-to-sequence learning substantially more powerful. Bahdanau, Cho, and Bengio introduced an attention mechanism that allowed the decoder to focus on different parts of the source sentence while generating the target sentence, rather than forcing the entire source

sentence through one fixed vector [43]. This was a major improvement inside Level 1. It did not change the regime into Level 3, because the model was still trained for a fixed translation task. But it made the supervised sequence mapping more flexible, more expressive, and more able to generalize across sentence pairs with complex alignments.

Google’s Neural Machine Translation system later showed that neural sequence-to-sequence models with attention, deep recurrent networks, residual connections, wordpiece modeling, beam search, and production-scale engineering could substantially reduce translation errors compared with Google’s previous phrase-based production system [44]. This mattered because it showed that neural NLP was not merely a research direction for small benchmarks. Like supervised CNNs in vision, neural sequence models could replace core parts of a traditional engineered pipeline in a major real-world language task.

Neural machine translation is especially important for the three-level framework because it clarifies the boundary between Level 1 and Level 3. NMT uses autoregressive decoding, but autoregression alone does not define Level 3. A translation model predicts target tokens, but it does so inside a fixed supervised conditional mapping from source sentence to target sentence. Its interface is translation, not open-ended reasoning, dialogue, coding, tool use, or general task behavior. Therefore, early NMT remains Level 1: a powerful supervised sequence generalizer. It later contributes mechanisms that become important for broader systems, but its original generalization regime is still task-bound.

The NLP side of Level 1 therefore adds an essential dimension to the story. Computer vision showed that deep networks could learn high-dimensional perceptual structure and defeat hand-engineered visual pipelines. NLP showed that deep networks could learn distributed linguistic structure, reduce reliance on sparse manual features, and perform structured sequence tasks such as translation. Both domains demonstrated the same first-level principle: neural networks could learn task-specific structures from data and generalize those structures to unseen samples.

3.5 What Level 1 Established

Level 1 established the first great regime of deep learning. It showed that neural networks could move beyond shallow pattern fitting and become strong supervised generalizers in difficult domains. In computer vision, this meant learning visual hierarchies from images rather than relying primarily on hand-designed descriptors. In NLP, it meant learning linguistic representations and sequence mappings rather than relying entirely on sparse features, phrase tables, and task-specific symbolic pipelines.

The key achievement of Level 1 was not generality in the later foundation-model sense. A supervised CNN did not become a general visual intelligence system. A neural machine translation model did not become a general language agent. These systems were still bound to fixed tasks. But within those tasks, they showed that deep models could internalize representation learning and generalize from finite supervised data to unseen samples at a level that changed entire fields.

This is the correct way to understand the first level. It is not “just supervised learning” in a dismissive sense. It is the stage in which deep learning first became practically powerful. It proved that large neural networks, when paired with enough data, compute, optimization, and appropriate architecture, could outperform feature-engineered systems in high-dimensional perception and language tasks. It also created the research culture that later made Level 2 and Level 3 possible: learn representations inside the model, scale the data and computation, reduce dependence on manual features, and observe what new form of generalization emerges.

At the same time, the boundary of Level 1 is clear. The model generalizes within a task, not across open-ended tasks. It may learn useful representations, but those representations are still mainly shaped by supervised objectives. It may use sequence prediction, but the prediction remains inside a task-

specific conditional mapping. Level 1 therefore made deep learning useful and credible, but it did not yet make it foundational in the representation-transfer sense or open-ended in the predictive-relational sense.

This boundary is exactly why Level 2 was necessary. Once deep learning proved that neural networks could learn strong task-specific structures, the next question became whether a model could learn reusable structure from data itself, before any particular downstream task. That transition—from supervised cross-sample generalization to self-supervised representation generalization—is the next level of the framework.

4. Level 2: Self-Supervised Representation Generalization

4.1 The Second Level Is a Change in the Object of Generalization

The second level of generalization is self-supervised representation generalization. This level begins when the central object of generalization is no longer a task-specific mapping from inputs to outputs, but a reusable representation learned from data itself. A Level 1 model is trained to solve a fixed task and to generalize that solution from seen samples to unseen samples. A Level 2 model is trained to learn internal structure that can be transferred across many downstream tasks. This is a genuine level transition, not a mere improvement in performance.

The difference is subtle but fundamental. In Level 1, the model is organized around a task. The training data contains labels or paired examples, and the model learns a function that performs that task on new samples. The representation may be useful, but it is mostly a means to an end. In Level 2, the representation becomes the central product of learning. The pretraining task is no longer primarily valuable because the model must perform that exact task later. It is valuable because it forces the model to extract structure from data in a form that can later support many different tasks.

For this reason, Level 2 should not be described simply as “better supervised learning” or “pretraining followed by fine-tuning.” Those phrases describe part of the procedure, but they miss the conceptual shift. The deeper point is that the model learns a reusable internal substrate. The downstream task does not have to build its entire structure from labeled examples. It can specialize a representation that has already absorbed broad regularities from unlabeled data.

This shift changes the meaning of generalization. At Level 1, generalization usually means that a classifier trained on one set of labeled examples performs well on unseen examples of the same task. At Level 2, the model also has to generalize from pretraining to downstream tasks. The model learns from one source of supervision, often produced automatically from the data, and the learned structure becomes effective in tasks whose labels, outputs, and evaluation formats may be very different from the pretraining objective.

The generalization object has therefore moved upward. The model is no longer only generalizing answers. It is generalizing representations. It is not only learning “how to classify these examples.” It is learning “what structure in this data can support many later uses.” This is the core of Level 2.

4.2 Why Self-Supervision Changes the Learning Regime

Self-supervision matters because it changes what the model can learn from. Supervised learning depends on human-provided labels, expert annotation, paired examples, or task-specific targets. These signals are powerful, but they are expensive and narrow. Most of the structure in the world is not explicitly labeled. Most language is not annotated with entailment labels, sentiment labels, named-entity tags, or question-answer spans. Most images are not labeled with every object, relation, part,

scene attribute, and semantic abstraction they contain. If a model can only learn from task labels, it can only access a small and human-selected part of the available structure.

Self-supervised learning turns data into its own training signal [24]. A model may predict missing words, reconstruct masked patches, contrast different views, align paired modalities, predict neighboring representations, or recover structure deliberately hidden from the input [2], [21]–[23], [45]. These objectives are not merely tricks for avoiding annotation. They create pressure for the model to learn regularities that are present in the data distribution itself. The label is no longer the only teacher. The structure of the data becomes the teacher.

This is the crucial transition from Level 1 to Level 2. Level 1 asks: given labeled examples of a task, can the model learn a function that works on unseen examples? Level 2 asks: given large amounts of raw data, can the model learn a representation that remains useful across many tasks? The second question is broader. It does not remove supervised learning, because downstream tasks often still use labels. But it changes the role of labels. Labels become a way to specialize a pretrained representation rather than the only source of learnable structure.

Level 2 is therefore closely related to the rise of foundation models. The foundation model report defines foundation models as models trained on broad data at scale and adaptable to a wide range of downstream tasks [8]. That definition captures the operational form of Level 2. A foundation-like representation model is not just a model that performs well on one dataset. It becomes a base on which many task-specific systems can be built. The model’s importance lies not only in one benchmark score, but in the breadth of downstream use made possible by the learned representation.

The word “representation” is therefore doing real work here. It does not merely mean a hidden vector inside a neural network. It means an internal organization of data regularities that can be reused. A strong representation compresses, separates, aligns, and structures information so that many later tasks become easier. It may encode syntax, semantics, visual parts, object identity, spatial relationships, cross-modal correspondences, or other latent regularities. The downstream task sees only a small slice of what the representation contains.

In this sense, Level 2 is the point at which deep learning begins to become infrastructure. A Level 1 model is often a product for a task. A Level 2 model is a base for many products. The same pretrained language encoder, visual backbone, or image-text representation model can be adapted, probed, fine-tuned, retrieved from, ranked with, or integrated into larger systems. The model is no longer only an answer machine for one problem. It is a reusable representational layer.

4.3 The Historical Need for Level 2

The transition to Level 2 was not accidental. It was driven by a deep limitation of Level 1. Task-specific supervised models can be very powerful, but they fragment learning. A separate dataset, objective, model head, and evaluation protocol are often built for each task. The model trained for one task may learn features that are useful elsewhere, but the training process is still organized around that task. This wastes structure, because the same underlying data often contains regularities useful for many tasks.

In NLP, this fragmentation was especially obvious. Language understanding tasks such as sentiment analysis, natural language inference, paraphrase detection, named entity recognition, question answering, semantic role labeling, and text classification all require overlapping linguistic knowledge. A model that understands syntax, lexical meaning, phrase composition, coreference, discourse, and commonsense relations should not have to relearn all of that separately for each dataset. Yet traditional pipelines and early task-specific neural systems often treated these tasks as largely separate engineering problems.

The deeper idea of Level 2 is that the common structure should be learned before the task. A model should first absorb broad regularities from raw data, then downstream tasks should specialize that structure. This is a more efficient and more powerful learning arrangement. It matches the fact that many tasks are different views of the same underlying world or language distribution. The task label is only one surface through which the deeper structure becomes visible.

This is also why Level 2 creates a different kind of scale law in practice. In Level 1, more data and more model capacity mainly improve the task-specific function. In Level 2, more data and more capacity can improve the representation itself, and that improvement may appear across many tasks. A stronger pretrained representation can lift performance in classification, retrieval, question answering, entailment, tagging, and other downstream settings at once. Scale no longer serves only one supervised mapping. It strengthens a shared substrate.

The historical significance of Level 2 is therefore larger than the success of any one model. It changes the workflow of deep learning. Instead of training every model from scratch on a narrow task, one begins with a pretrained representation. Instead of treating each downstream dataset as the full source of knowledge, one treats it as a way to adapt broad pretrained structure. Instead of building all intelligence into a task-specific pipeline, one lets the model learn much of the structure from data itself.

This does not mean Level 2 is already Level 3. A representation model may be extremely powerful without becoming an open-ended generator or reasoning system. BERT is the canonical example. It learns strong contextual representations from text and transfers them across many tasks, but its main interface is still representation and fine-tuning, not open-ended generation. This boundary is important. Level 2 is more general than Level 1, but it is not yet the predictive high-dimensional relational regime of Level 3.

4.4 NLP as the First Clear Crystallization of Level 2

NLP was the first domain where Level 2 became unmistakably visible. This happened because language offers an enormous amount of raw data and because many NLP tasks share deep structure. A model that learns useful structure from unlabeled text can later help with natural language inference, question answering, sentiment analysis, paraphrase detection, named entity recognition, and many other tasks. The same data distribution contains the material for many downstream capabilities.

The road to Level 2 in NLP began before BERT. The neural probabilistic language model had already proposed learning distributed word representations as part of a probability model over word sequences [38]. Word2vec later made distributed word representations practical and influential at scale, showing that word and phrase vectors learned from large corpora could capture syntactic and semantic regularities [46]. These models did not yet fully instantiate BERT-style contextual representation pretraining, but they helped establish the key principle that linguistic structure could be learned as continuous representation rather than manually engineered as sparse symbolic features.

The limitation of static word vectors was that a word type received one representation even though meaning depends on context. This made contextual representation learning the natural next step. ELMo introduced deep contextualized word representations learned from the internal states of a pretrained bidirectional language model and showed that these representations could improve a broad range of NLP tasks [47]. ULMFiT then showed that language model pretraining and fine-tuning could serve as a general transfer learning method for text classification, rather than a task-specific trick [48]. These models were transitional because they made the Level 2 logic increasingly clear: learn from raw text first, then adapt to supervised tasks.

BERT crystallized the transition. It did not merely add another useful embedding method. It made self-supervised representation generalization the dominant NLP paradigm. BERT pretrained a deep

bidirectional Transformer encoder on unlabeled text using masked language modeling and next sentence prediction, then fine-tuned the same pretrained model across many downstream tasks with minimal task-specific architectural change [2]. This is exactly the Level 2 form: a broad self-supervised pretraining process produces a reusable representation, and downstream tasks specialize it.

The architecture mattered, but it was not the whole point. BERT used the Transformer encoder, which made deep contextual representation learning scalable through self-attention [49]. Its masked language modeling objective allowed the model to condition on both left and right context, making it especially suitable for language understanding tasks [2]. But the historical importance of BERT was not just “Transformer plus masking.” Its deeper importance was that it gave NLP a general representation base. The model was no longer built separately for one task. The same pretrained encoder could be fine-tuned for natural language inference, question answering, sentence-pair classification, sequence labeling, and more.

The performance results made this shift undeniable. BERT achieved state-of-the-art results on eleven NLP tasks, including GLUE, MultiNLI, and SQuAD [2]. GLUE was designed as a benchmark for general language understanding across multiple NLU tasks rather than a single isolated dataset [50]. SQuAD tested reading comprehension through answer span extraction, and SQuAD 2.0 added unanswerable questions that required models to decide when a passage did not contain an answer [51]. BERT’s strong results across these different formats showed that the value of the model was not confined to one task head. Its value was the transferable representation learned during pretraining.

This is the crucial point: BERT was not simply a better supervised NLP model. It changed what counted as the main model. Before BERT-style pretraining, a downstream task model often carried most of the modeling burden. After BERT, the pretrained encoder became the main carrier of linguistic structure, and the downstream task became a specialization layer. This changed the default workflow of NLP research and engineering. The question was no longer only “what architecture should we design for this task?” It became “how should we adapt the pretrained representation to this task?”

BERT is therefore the central NLP landmark of Level 2. It made representation transfer the default logic of modern NLP. It showed that a model trained on raw text with a self-supervised objective could become a general-purpose language representation substrate. It did not make NLP open-ended in the GPT sense, but it made NLP foundational in the representation sense.

4.5 Visual and Multimodal Expansion of Level 2

NLP made Level 2 unmistakable, but the same generalization regime was not limited to language. Computer vision also had to move beyond Level 1 because supervised labels were too narrow to exhaust visual structure. ImageNet-style labels were extremely useful, but they compressed an image into a fixed category space [1], [33]. A single label such as “dog” or “airplane” does not fully describe object parts, spatial layout, texture, pose, scene relations, material properties, affordances, or broader semantic context. The label can organize a strong classifier, but it cannot serve as the whole structure of vision.

Visual self-supervised learning therefore asked a different question: can images teach models without human labels? Early pretext tasks made this question concrete. Context prediction trained a model to infer the spatial relationship between image patches [52]. Rotation prediction trained a model to identify the rotation applied to an image [53]. These tasks were not important because patch position or rotation was the final downstream goal. They were important because they showed that images contain internal supervisory signals. A model can be forced to learn useful visual structure by solving tasks generated from the data itself.

The early pretext-task phase was necessary but limited. Many such tasks could be solved partly through low-level cues, local statistics, shortcuts, or task-specific artifacts. This meant they did not yet establish a mature general visual representation regime. They proved that visual self-supervision was possible, but they did not yet produce the kind of large-scale, transferable representation breakthrough that BERT produced in NLP. The field needed objectives that shaped the representation space more directly.

Contrastive learning became the major turning point. Its central move was to train the model not to predict a human label, but to organize relationships among views. Two augmented views of the same image should be close in representation space; unrelated images should be separated. The model learns invariance, identity preservation, and semantic separation through the geometry of representations rather than through category labels. Contrastive learning is therefore important for Level 2: the target is not a task answer, but a representation space.

CPC was one of the key roots of this transition. The original idea of Contrastive Predictive Coding framed representation learning as contrastive prediction in latent space. Its improved image-recognition version, often referred to as CPC v2, was one of the first major ImageNet-scale performance breakthroughs in contrastive visual self-supervision [45]. CPC v2 reported a new state of the art in ImageNet linear classification with 71.5% top-1 accuracy, compared with 48.7% for the original CPC implementation [45]. It also showed strong data-efficiency: when a downstream classifier was trained on CPC features, the system reached 83.4% top-1 and 96.5% top-5 accuracy with the full ImageNet labels, surpassing a supervised ResNet-200 baseline in that specific downstream setup [45]. This second result should not be confused with linear probing, but it is historically important because it showed that self-supervised visual representations could become extremely strong inputs to supervised recognition systems.

MoCo then made contrastive visual representation learning more general and more practical. It treated contrastive learning as dictionary lookup and introduced a dynamic dictionary maintained by a queue and a momentum encoder [54]. The key problem was that contrastive learning often benefits from many negative examples, but very large comparison sets are expensive. MoCo solved this by maintaining a large and consistent dictionary without requiring an enormous batch size. Its original ResNet-50 model achieved 60.6% ImageNet linear accuracy, and wider variants reached 68.6%; the paper also notes that MoCo v2 reached 71.1% with a ResNet-50 through improved augmentations and projection-head changes [54]. MoCo’s deeper significance was not only accuracy. It made contrastive learning look like a scalable and transferable representation-learning framework rather than a specialized pretext trick.

SimCLR then made the contrastive paradigm cleaner and more mature. It removed specialized mechanisms such as memory banks and focused on a simple framework: two augmented views, a contrastive loss, strong data augmentation, a projection head, large batches, and longer training [21]. SimCLR’s contribution was partly empirical and partly conceptual. It showed that the choice of augmentation is not a detail; it defines what invariances the model is forced to learn. It also showed that a clean contrastive recipe could scale to very strong results. A linear classifier trained on SimCLR representations achieved 76.5% ImageNet top-1 accuracy, matching supervised ResNet-50 performance; with only 1% of labels, fine-tuning achieved 85.8% top-5 accuracy, outperforming AlexNet with 100 times fewer labels [21]. SimCLR was therefore not the first contrastive ImageNet breakthrough, but it was one of the clearest moments when contrastive learning became a simple, scalable, high-performance visual representation paradigm.

DINO marked another shift. It showed that visual self-supervision was not only about instance discrimination or contrastive separation. DINO used self-distillation without labels, training a student network to match a momentum teacher, and revealed striking emergent properties in self-supervised Vision Transformers [55]. The key result was not merely its 80.1% top-1 ImageNet linear evaluation

with ViT-Base. More important was the qualitative structure of the representation: self-supervised ViT features contained explicit information about semantic segmentation, and the attention maps often highlighted meaningful object regions even though no segmentation labels were used [55]. This was a crucial Level 2 signal. The representation was not just good for a downstream classifier; it appeared to organize visual scenes in a semantically meaningful way beyond the explicit pretraining target.

MAE then made the visual Level 2 regime look simpler and more scalable. Masked autoencoders trained a Vision Transformer by masking a high proportion of image patches, such as 75%, and reconstructing the missing pixels [22]. The key design was an asymmetric encoder-decoder: the encoder processed only visible patches, while a lightweight decoder reconstructed the missing content [22]. This was not just a technical trick. It made masked visual pretraining efficient and scalable for large Vision Transformers. MAE showed that a model could learn strong visual representations by reconstructing missing structure from partial observation, and a vanilla ViT-Huge trained with MAE achieved 87.8% ImageNet accuracy using only ImageNet-1K data [22]. In the framework of this paper, MAE matters because it made masked reconstruction a serious scalable route to visual representation generalization.

CLIP extends Level 2 across modality boundaries. Its central achievement is not merely image-text retrieval. It changes the source and interface of visual representation learning. Instead of learning from a fixed set of human-provided category labels, CLIP learns from natural language supervision over large-scale image-text pairs [23]. The model trains an image encoder and a text encoder so that matching image-text pairs are aligned in a shared representation space. This gives the model a much broader semantic interface than ImageNet labels: new visual concepts can be specified in language rather than learned through a new supervised dataset.

CLIP is a strong form of Level 2 because its central product is representation alignment. A supervised ImageNet classifier generalizes within a fixed label set. SimCLR, DINO, and MAE learn visual representations that transfer across tasks. CLIP goes further by aligning visual and linguistic representations, enabling zero-shot transfer through natural language prompts. It was trained on 400 million image-text pairs and evaluated over more than 30 datasets; it matched the original ResNet-50 on ImageNet zero-shot without using the 1.28 million labeled ImageNet training examples [23]. CLIP is not Level 3, because it is not primarily an open-ended reasoning or acting system. But within Level 2, it is a major expansion: representation generalization becomes multimodal.

Together, these models show that Level 2 is not a BERT-only phenomenon. In NLP, self-supervision learns transferable language representations. In vision, contrast, self-distillation, and masked reconstruction learn transferable visual representations. In multimodality, image-text alignment learns representations that cross modality boundaries. The methods differ, but the generalization regime is the same: the model learns structure from data-derived supervision, and the learned representation becomes useful beyond the original pretraining objective.

4.6 The Scale Logic of Level 2

Scale continues to matter in Level 2, but it acts on a different object than in Level 1. In Level 1, scale strengthens a task-specific supervised mapping. In Level 2, scale strengthens a reusable representation. This is the central distinction. A larger supervised classifier may become better at one label space. A larger pretrained representation model may become a stronger base for many tasks, many datasets, or even multiple modalities.

BERT made this logic clear in NLP. BERT Base had 12 Transformer layers, 768 hidden size, 12 attention heads, and about 110 million parameters; BERT Large had 24 layers, 1024 hidden size, 16 attention heads, and about 340 million parameters [2]. These models were pretrained on BooksCorpus

and English Wikipedia, then fine-tuned across downstream tasks [2]. The larger model did not merely improve one classifier. It improved the shared linguistic representation that downstream tasks could specialize.

RoBERTa sharpened the lesson. Liu et al. argued that BERT had been undertrained and showed that performance could improve through longer training, larger mini-batches, more data, dynamic masking, and removing next sentence prediction [56]. The meaning of RoBERTa is not that BERT was conceptually wrong. The meaning is that the BERT-style Level 2 regime had substantial headroom once data scale, compute, and optimization were pushed harder. Scale was not only making a model bigger. It was making the representation substrate stronger.

The visual side shows the same principle through a different path. CPC v2 showed that a stronger contrastive representation could make ImageNet recognition more data-efficient and reach a new linear-probe state of the art [45]. MoCo showed that scaling the comparison structure through a queue and momentum encoder made contrastive learning practical and transferable [54]. SimCLR showed that clean contrastive learning benefits from large batches, strong augmentations, projection heads, and longer training [21]. DINO showed that self-supervised ViTs could develop semantic structure not directly supplied by labels [55]. MAE showed that masked visual pretraining could scale efficiently to large Vision Transformers [22]. CLIP showed that representation learning can scale across hundreds of millions of image-text pairs and produce a flexible multimodal interface [23].

The shared pattern is the same as the AlexNet lesson, but at a higher level. AlexNet proved that deep networks could enter a new supervised generalization regime when model scale, data scale, GPU compute, and training technique aligned. BERT and RoBERTa showed that language representation learning could enter a new regime when large unlabeled corpora, Transformer encoders, self-supervised objectives, and fine-tuning aligned. CPC v2, MoCo, SimCLR, DINO, MAE, and CLIP showed that visual and multimodal representation learning could also be scaled into stronger regimes when the objective, architecture, data, and compute were aligned.

Level 2 is therefore closely related to the foundation-model layer of the framework. Foundation models are trained on broad data at scale and adapted to a wide range of downstream tasks [8]. In this paper, however, the term foundation model is an operational category, whereas Level 2 and Level 3 are generalization-regime categories: some foundation models are Level 2-dominant, while GPT-style foundation models are Level 3-dominant. Level 2 explains the generalization mechanism behind the representation-centered part of that operational definition. The model becomes foundational because the representation becomes broadly reusable. Scale does not merely increase task accuracy. It increases the amount and quality of structure stored in the representation.

The scale logic of Level 2 is therefore not simply “bigger is better.” It is more precise: when the pretraining objective forces the right kind of structure, and when the model, data, and compute are large enough, the learned representation becomes useful beyond the task or signal that produced it. That is the distinctive form of Level 2 scale.

4.7 The Boundary of Level 2

Level 2 is a major expansion beyond Level 1, but it has a clear boundary. A Level 2 model learns representations that transfer across tasks, datasets, or modalities, but it usually still requires a downstream mechanism to turn representation into behavior. That mechanism may be fine-tuning, a task head, a span extractor, a classifier, a retrieval system, a reranker, a segmentation head, a detection pipeline, a zero-shot prompt template, or a larger system built around the representation. The model’s capability is primarily exposed through adaptation, not through a single open-ended interface.

This boundary is easiest to see in BERT. BERT is enormously important because it gives many NLP systems a strong language representation base. It can support classification, question answering,

retrieval, ranking, tagging, and document understanding. But it is not normally used by asking it arbitrary questions and receiving long generated answers. Its main interface is not free-form generation. Its power lies in representation and adaptation.

The same boundary applies to visual and multimodal Level 2. SimCLR, DINO, and MAE can produce strong visual representations, but those representations usually become useful through linear probing, fine-tuning, detection heads, segmentation systems, retrieval systems, or transfer pipelines. CLIP is more flexible because natural language can define new visual categories at test time, but its core is still image-text representation alignment. It compares representations; it does not by itself become an open-ended agent, planner, coder, or reasoning system.

This is not a weakness. It is the defining power of Level 2. The second level makes deep learning foundational by producing reusable representation infrastructure. It turns models into bases on which many downstream systems can be built. It allows one pretraining process to support many tasks. It reduces dependence on task-specific feature engineering and narrow labeled datasets. It makes deep learning less like a collection of isolated task solvers and more like a layered infrastructure of learned structure.

But Level 2 is not the end of the story. A strong representation base does not automatically become an open-ended intelligence interface. It may encode a large amount of useful structure, but that structure still has to be released through some downstream mechanism. For this reason, BERT and GPT should not be collapsed into one simple category. BERT is the mature NLP form of Level 2: a pretrained encoder whose main value is transferable representation. GPT begins the path toward Level 3: an autoregressive predictive model whose interface later becomes open-ended generation, instruction following, coding, tool use, and reasoning.

The conclusion of Section 4 is therefore clear. Level 1 generalizes task-specific mappings across unseen samples. Level 2 generalizes learned representations across tasks, datasets, and modalities. NLP provided the clearest early crystallization through BERT. Computer vision matured through contrastive learning, self-distillation, and masked reconstruction. Multimodality expanded the regime through image-text alignment. Together, these developments established the second major level of deep learning generalization: from task-specific mappings to reusable representations.

5. Level 3: Autoregressive and Predictive High-Dimensional Relational Generalization

5.1 The Third Level Is Not Just Generation, but Predictive Relational Generalization

The third level of generalization is autoregressive and predictive high-dimensional relational generalization. This level begins when the central object of generalization is no longer only a task-specific mapping, as in Level 1, or a reusable representation, as in Level 2, but a predictive structure over high-dimensional relations. The model is trained to predict the next token, the next state, the missing part, the future representation, the next action consequence, or the future dynamics of a data stream. Through that predictive training, the model may learn relational structure rich enough that its output process itself becomes a general interface for task behavior [4], [5], [15], [16], [57].

This is a genuine level transition. Level 1 asks whether a model can learn a supervised mapping and generalize that mapping from seen samples to unseen samples. A supervised image classifier maps an image to a label. A neural machine translation system maps a source sentence to a target sentence. A sentiment model maps a text to a sentiment class. The task is fixed, and the model generalizes within that task [42].

Level 2 asks whether a model can learn a reusable representation from data itself and transfer that representation across tasks. A BERT-like encoder learns contextual language representations from unlabeled text, then downstream systems specialize those representations through fine-tuning, classification heads, span extraction heads, retrieval systems, ranking systems, or probing. The model becomes a representation base. Its central product is not one answer, but a reusable internal structure [2], [8].

Level 3 asks a different question: can predictive training over a rich data stream produce a model whose output stream can directly function as the task interface? In this regime, the task does not always need to be converted into a separate supervised dataset or downstream head. The task can be written into the context. The model reads the context, continues the sequence, and the continuation can be the answer, explanation, proof, code, plan, tool call, dialogue response, or reasoning trace. The model is not merely a representation waiting to be adapted. Its prediction process is itself the way capabilities are released [4], [5], [19], [57], [58].

Level 3 should therefore not be reduced to “generative AI” in a shallow sense. Many models generate outputs. A translation model generates a target sentence. An image model generates pixels. A small text model generates words. But generation alone does not define Level 3. The deeper question is what role prediction plays in the system. If prediction is only a narrow task output, the model may remain Level 1. If prediction is only a pretraining pressure for learning representations, the model may remain Level 2. Level 3 begins when prediction becomes a broad interface through which learned relational structure can be used across many task contexts [2], [4], [5], [19], [42], [57].

The decisive example is the GPT-style language model. The model is trained to predict the next token in broad text and code distributions. At first glance, this looks simple. But language and code are not ordinary sequences. They are dense external records of human symbolic activity. Text contains explanations, questions, arguments, instructions, plans, social interaction, scientific reasoning, mathematical derivations, legal reasoning, documentation, and everyday problem solving. Code contains executable procedures, abstractions, dependencies, decomposition, and tool-like reasoning. A model trained at large scale to predict such data is not only learning word co-occurrence. It is pressured to learn a large amount of the relational structure externalized in human symbolic activity [4], [5], [9], [57], [59], [60].

Level 3 should also not be reduced to “LLMs imitate human behavior.” That description is too weak. Large language models do not merely copy surface behavior in the trivial sense. At sufficient scale, language modeling over broad text, code, symbols, and interaction traces forces the model to learn a powerful subset of human-intelligence-like structure externalized in language. This structure includes patterns of abstraction, explanation, analogy, problem decomposition, planning, proof, programming, instruction following, dialogue, tool use, and step-by-step reasoning. It is not the full structure of human intelligence. It is not full AGI. But it is also not shallow imitation [5], [9], [10], [60], [61].

The core claim is therefore precise: LLMs have learned a powerful subset of human-intelligence-like structure externalized in language, code, symbols, and textual interaction, and this structure can be released as both open-ended task behavior and strong human-like reasoning capability. This is the first decisive realization of Level 3 [5], [9]–[12], [60], [61].

5.2 Why Not Every Predictive Task Is Level 3

The hardest and most important distinction is this: prediction alone does not create Level 3. Many deep learning systems predict something. A translation model predicts target tokens. BERT predicts masked tokens. A classifier predicts a label. A masked autoencoder predicts missing pixels. A time-series model predicts future values. But these systems do not automatically belong to the same

generalization level. The level depends on the role prediction plays in the model’s training, output interface, and capability release [2], [4], [5], [22], [42], [57].

The clearest comparison is among three cases: neural machine translation, BERT, and GPT.

A neural machine translation model is predictive, but its prediction is locked inside a fixed conditional task. In a sequence-to-sequence translation model, the input is a source sentence, such as an English sentence. The encoder reads the source sentence and compresses it into hidden states or a vector-like representation. The decoder then generates the target sentence, such as the French translation, one token at a time. Each target token is predicted conditioned on the source sentence and the previous target tokens. The model’s output is a translation. Its interface is not open-ended. It cannot naturally be asked to solve arbitrary tasks by describing them in the input. It is trained and evaluated as a source-to-target mapping [42].

Neural machine translation therefore remains Level 1, even though it uses autoregressive decoding. The prediction is real, but it serves a fixed supervised mapping. The training data consists of paired source-target examples. The objective is to map unseen source sentences to target translations. The model generalizes across sentence pairs, not across arbitrary task contexts. Autoregression is used as a mechanism inside the task, not as a general task interface. A decoder that predicts French tokens from an English sentence is not the same kind of system as a model that can receive a natural language task description, examples, constraints, and context, then generate the relevant answer or reasoning path [42], [43].

BERT is also predictive, but in a different way. During pretraining, BERT receives text with some tokens masked out. The model uses both left and right context to predict the original masked tokens. For example, from a sentence containing [MASK], it produces a probability distribution over the vocabulary for that masked position. It also uses next sentence prediction in the original formulation. This prediction is important, but it is not the final capability interface. BERT is not mainly used by giving it an arbitrary prompt and asking it to continue the text. After pretraining, BERT is usually used as an encoder. Its hidden states are passed to a downstream classifier, span extractor, token tagger, retrieval module, or other task head. The learned representation is the product; the masked prediction objective is the training pressure [2].

BERT is therefore Level 2. The model learns bidirectional contextual representations from unlabeled text, and those representations are adapted to downstream tasks. Its pretraining prediction is not itself an open-ended behavior interface. It is a way to force the encoder to learn useful structure. In downstream use, the task is usually not solved by BERT freely generating a continuation. The task is solved by attaching or fine-tuning a mechanism that reads BERT’s representations [2], [19].

GPT-style models are different in the most important possible way. A GPT model receives a text context and predicts the next token. Then the predicted token becomes part of the context, and the model predicts the next token again. This recursive continuation process is not restricted to translation, masked-token recovery, or one predefined output format. The context can contain a question, an instruction, a few demonstrations, a code file, a mathematical problem, a dialogue history, a tool specification, a scientific paragraph, a legal argument, or a plan. The output can be an answer, a summary, a proof, a program, an explanation, a chain of reasoning, a function call, or a next action description. The prediction interface is flexible enough to carry the task itself [4], [5], [9], [19], [57].

This is the core distinction. In NMT, the input defines a translation mapping. In BERT, the input supports representation extraction. In GPT, the input context can define the task, the format, the constraints, the examples, and the interaction history. The output is not merely a label or a representation. The output is the model’s behavior in the task [2], [4], [5], [19], [42], [57].

GPT-2 made this difference visible. Its central claim was that language models are unsupervised multitask learners: by training on a large WebText corpus and using a natural language context, a language model could perform tasks such as translation, summarization, question answering, and reading comprehension in a zero-shot setting [4]. The model was not fine-tuned separately for each task. The task appeared in the context, and the model produced a continuation. This was not yet the full modern assistant interface, but it was the beginning of a new regime [4], [19].

GPT-3 made the difference undeniable. GPT-3 was a 175-billion-parameter autoregressive language model evaluated in zero-shot, one-shot, and few-shot settings without gradient updates or task-specific fine-tuning [5]. Tasks and demonstrations were specified purely through text interaction with the model [5]. This is the decisive Level 3 form. The model does not require a downstream task head to release its capability. The context becomes the task interface. The continuation becomes the task output. The same next-token prediction mechanism can produce translation, arithmetic attempts, analogies, code, explanations, classification answers, and question answering behavior [5], [9], [19].

Level 3 prediction is special because it is not prediction as a narrow target. It is prediction as a universal interface. The model predicts tokens, but because the context can describe tasks and the output can instantiate solutions, next-token prediction becomes a mechanism for open-ended task behavior. The predictive objective trains the model; the predictive interface releases the model’s learned relational structure [4], [5], [9], [19], [57], [60], [61].

5.3 BERT and GPT Are Different Levels, Not Merely Different Architectures

The BERT-GPT distinction is so important that it deserves a direct statement even before the later dedicated section. BERT and GPT are not merely two Transformer variants. They represent different generalization regimes [2], [4], [5], [19], [57].

BERT is an encoder-centered representation model. Its pretraining objective is designed to learn deep bidirectional representations from unlabeled text by conditioning on both left and right context [2]. During use, BERT usually consumes an input sequence and produces contextual hidden states. A task-specific mechanism then uses those hidden states: a classifier reads the [CLS] representation, a span extractor selects start and end positions, a token classifier labels each token, or a retrieval system uses embeddings. The model’s power is real, but it is released through representation adaptation [2].

GPT is a decoder-centered autoregressive predictive model. Its pretraining objective is to predict the next token from previous context. During use, the same mechanism continues the sequence. There is no fundamental separation between “pretraining objective” and “inference interface.” The model was trained to continue text, and it is used by asking it to continue text. Once tasks can be written as text, the model’s output stream becomes a task-solving interface [4], [5], [19], [57].

This difference is not cosmetic. It changes how generalization works. BERT generalizes by producing representations that downstream systems can reuse. GPT generalizes by using context to define new tasks and generating the appropriate continuation. BERT requires adaptation mechanisms for most tasks. GPT can perform many tasks through prompting and in-context examples without gradient updates, as GPT-3 showed [5]. BERT’s learned structure is mainly accessed through hidden states. GPT’s learned structure is accessed through the token stream [2], [5], [19].

BERT is therefore Level 2 and GPT is Level 3. BERT’s central achievement is self-supervised representation generalization. It learns reusable representations from text and transfers them across tasks. GPT’s central achievement is predictive relational generalization. It learns from broad symbolic sequences and releases learned structure through autoregressive continuation. Both are self-supervised in a broad sense. Both are Transformer-based. Both can be large. But their capability interfaces are different, and that difference places them in different levels of this framework [2], [4], [5], [19], [57].

The distinction can be made concrete:

A BERT-style model sees an input such as: “The capital of France is [MASK].” During pretraining, it predicts the missing token. Later, for a task such as sentiment classification, the model reads a sentence and a classifier head maps its representation to a label. The model does not naturally produce a long answer, explain its reasoning, write code, or follow arbitrary instructions through the same interface [2].

A GPT-style model sees an input such as: “Question: What is the capital of France? Answer:” and continues with “Paris.” The same interface can accept: “Translate this sentence into French,” “Write a Python function,” “Explain this theorem,” “Classify the sentiment,” “Solve this equation step by step,” or “Given these examples, infer the pattern.” The task is embedded in the prompt. The output is generated directly. This is not merely a different architecture. It is a different relationship between training, inference, and task behavior [4], [5], [9], [19], [57].

5.4 Why Language Modeling Became the First Successful Level 3 Path

Language modeling became the first successful path to Level 3 because language is not just data. It is a compressed externalization of human cognitive activity. Human beings use language to explain, ask, answer, instruct, argue, plan, imagine, teach, prove, document, persuade, negotiate, and coordinate. Code adds an even more structured layer: it records procedures, abstractions, modular decomposition, formal constraints, execution logic, and tool-like operations [4], [5], [9], [57], [60].

This means that next-token prediction over broad language and code distributions is not like predicting the next symbol in a random string. The model is trained on a distribution filled with human-generated relational structure. To reduce prediction error, the model must increasingly capture regularities of grammar, meaning, factual association, discourse structure, instruction structure, code dependency, mathematical notation, rhetorical pattern, and problem-solving trace. The larger and broader the training distribution, the more pressure there is to learn relations that go beyond local word statistics [5], [9], [57], [59], [60].

At small scale, a language model may produce mostly shallow completion. It may imitate style, repeat phrases, or exploit local correlations. But at large scale, the task becomes much harder and richer. Predicting the next token in a proof may require tracking assumptions and conclusions. Predicting code may require tracking variable dependencies and function behavior. Predicting an answer in a tutorial may require modeling the instructional relation between question and explanation. Predicting a chain of reasoning may require following intermediate transformations. The same formal training objective remains next-token prediction, but the structure required to perform it well becomes deeper [5], [9], [10], [57], [60]–[62].

The Level 3 claim is therefore not magical. It does not say that intelligence appears from nothing. It says that broad predictive training over a dense symbolic record can force the model to internalize many useful relational structures. Language and code contain traces of human reasoning, but not in a clean labeled form. Autoregressive prediction is the mechanism that turns those traces into a learning signal [4], [5], [9], [10], [57], [59].

Scaling laws made this path more credible. Kaplan et al. showed that language model performance scales predictably with model size, data size, and compute across many orders of magnitude [57]. This mattered because it suggested that autoregressive language modeling was not merely a one-off trick. It was a scalable learning regime. If predictive loss improves with scale, and if downstream behavior emerges from better predictive modeling, then scaling the predictive model becomes a route to broader capability [5], [57], [60], [61].

GPT-2 and GPT-3 provided the empirical turning points. GPT-2 showed that language models begin to perform tasks without explicit task-specific supervision when trained on large web text [4]. GPT-3 showed that scaling this regime to 175 billion parameters produced strong zero-shot, one-shot, and few-shot behavior across many tasks, with tasks specified through text interaction rather than gradient updates [5]. GPT-style models are therefore the first decisive Level 3 systems. They turned language prediction into a general task interface [4], [5], [19], [57].

5.5 From Predictive Modeling to Instruction Following

Raw predictive modeling is not enough to produce a reliable assistant. A model can learn enormous relational structure from text and code but still produce outputs that are unhelpful, untruthful, unsafe, evasive, verbose, miscalibrated, or misaligned with the user’s intention. The transition from GPT-3-style language modeling to instruction-following systems is therefore not a side detail. It is a necessary step in making Level 3 usable [5], [62].

InstructGPT showed this clearly. Ouyang et al. argued that making language models larger does not inherently make them better at following user intent, and they trained models using supervised demonstrations and reinforcement learning from human feedback [62]. Human evaluators preferred InstructGPT outputs to outputs from much larger GPT-3 models, even when the InstructGPT model had far fewer parameters [62]. This result matters because it separates two parts of Level 3: pretraining builds broad predictive relational structure, while instruction tuning and preference learning shape how that structure is released to users [62].

ChatGPT then made the interface public and conversational. OpenAI described ChatGPT as a sibling model to InstructGPT, trained to follow instructions in a prompt and provide detailed responses; its dialogue format made it possible to answer follow-up questions, admit mistakes, challenge incorrect premises, and reject inappropriate requests [63]. Historically, this was important because it transformed the autoregressive predictive model from a research object into an everyday interactive system. The model became not only a language model, but an interface [63].

This step is essential for understanding Level 3. Open-ended capability requires more than prediction quality. It requires controllability. A model may contain large amounts of learned relational structure, but users need a way to direct that structure toward tasks. Instruction following, dialogue formatting, preference tuning, safety behavior, and refusal behavior are not the same as intelligence itself, but they determine how the intelligence-like structure becomes usable. They turn predictive continuation into task-oriented interaction [62], [63].

This also explains why Level 3 is layered. Pretraining creates the base predictive model. Instruction tuning aligns the model with task requests. Reinforcement learning from human feedback or other preference signals shapes output quality and behavior. Chain-of-thought prompting and reasoning training organize intermediate computation. Tool use extends the model’s predictions into external operations. Retrieval and memory systems provide additional context. These layers do not replace the Level 3 base; they make its predictive interface more useful [10]–[12], [19], [58], [62], [63].

5.6 Reasoning as Predictive Structure Released Through Tokens

The strongest evidence that Level 3 is not merely surface imitation comes from reasoning. Large language models can generate intermediate reasoning steps, manipulate symbols, solve mathematical problems, write and debug code, follow abstract instructions, and adapt to new task formats in context. These abilities are imperfect and sometimes brittle, but they are real enough that any theory of modern deep learning must explain them [5], [9], [10], [60], [61].

Chain-of-thought prompting was an early clear signal. Wei et al. showed that prompting large language models to generate intermediate reasoning steps significantly improves performance on

arithmetic, commonsense, and symbolic reasoning tasks [10]. The important point is not only the prompting method. The deeper point is that the model’s predictive stream can be organized into intermediate structures that function as reasoning. The model does not merely output a label. It can produce a path from problem to answer [10].

This is a major Level 3 phenomenon. In Level 1, reasoning-like structure usually lives outside the model, inside the task design or external pipeline. In Level 2, reasoning-relevant information may be latent inside representations, but the primary interface is still representation adaptation. In Level 3, reasoning can appear inside the generated token stream itself. The model can write the decomposition, intermediate calculation, proof sketch, comparison, explanation, code, or plan. Prediction becomes a medium for reasoning behavior [9]–[12], [60], [61].

Reasoning-focused models made this even more explicit. OpenAI described o1 as a model series designed to spend more time thinking before responding, with stronger performance on difficult math, science, and coding tasks than previous models [11]. DeepSeek-R1 explored reasoning capability through reinforcement learning, reporting that DeepSeek-R1-Zero developed behaviors such as self-reflection, verification, and dynamic strategy adaptation, while DeepSeek-R1 used multi-stage training to improve performance and readability [12]. These systems suggest that reasoning is not only a byproduct of ordinary language modeling. It can be further incentivized, shaped, and organized on top of the predictive model base [11], [12].

This does not mean LLM reasoning is identical to human reasoning. It is not. Humans have embodiment, perception, affect, memory, agency, biological drives, social grounding, and developmental experience that current LLMs do not possess in the same form. But it also does not mean LLM reasoning is trivial imitation. The model has learned patterns of symbolic transformation, abstraction, explanation, verification, and problem decomposition from enormous traces of human symbolic activity. When these patterns are organized through prompting, instruction tuning, reinforcement learning, and test-time computation, they can produce strong human-like reasoning behavior [5], [9]–[12], [60], [61].

Level 3 must therefore be described carefully. LLMs are not full human minds. They are not complete agents. They do not possess the full structure of embodied intelligence or autonomous world modeling. But they have learned a powerful subset of human-intelligence-like structure externalized in language and code. That subset can be released as task behavior and reasoning. This is one of the most important facts in the history of deep learning [5], [9]–[12], [60], [61].

5.7 Tool Use, Actions, and the Expansion of the Predictive Interface

Once the output stream becomes a task interface, it can also become an action interface. This is another reason Level 3 is different from Level 2. A representation model may be used inside a tool pipeline, but it does not naturally produce tool-use sequences as its main behavior. A GPT-style model can generate text that represents actions: API calls, search queries, code execution steps, calculator calls, database operations, calendar actions, and other structured outputs [9], [58].

Toolformer made this idea explicit. It showed that language models can teach themselves to use external tools through simple APIs, deciding which APIs to call, when to call them, what arguments to pass, and how to incorporate the results into future token prediction [58]. This matters because it shows that predictive modeling can be extended beyond ordinary text into structured interaction with external systems. The model predicts not only words, but action-like tokens that can call tools and change the information available for future prediction [58].

This does not make every tool-using system a complete agent. Tool use can be shallow, brittle, scaffolded, or heavily constrained. But it shows the direction of Level 3. The predictive interface can be connected to external computation. Code generation is another example. When a model writes

code, the output is not merely natural language. It is an executable structure. If the code is run, tested, corrected, and iterated, the model’s token predictions become connected to computation in the world [9], [58].

Level 3 therefore naturally expands into agents, tool-using systems, coding assistants, web-browsing systems, and workflow automation. The base capability is still predictive relational generalization. But the output stream can be interpreted as commands, programs, plans, or actions. This is a different kind of interface from a BERT embedding or a CLIP representation. The model’s generated sequence can directly participate in a task loop [9], [58].

5.8 Why Level 3 Is Broader Than LLMs

Although GPT-style LLMs are the first successful Level 3 systems, Level 3 should not be defined as “large language models.” That would be too narrow. The correct definition is broader: a Level 3 system learns high-dimensional relational structure through autoregressive or predictive training, and its predictive interface can support open-ended behavior, reasoning, planning, simulation, action, or world modeling [5], [15], [16].

Language was the first successful path because it is a dense external record of human symbolic activity. But future Level 3 systems may learn through video, multimodal interaction, robotics, action-conditioned prediction, latent world states, or representation-space prediction. The predictive target may be the next token, but it may also be the next video patch, the next latent state, the future representation of a scene, the consequence of an action, or the hidden structure of a world process [15], [16].

This broader view prevents the framework from being trapped inside current LLMs. Video generation models, for example, can be interpreted as attempts to learn predictive structure over visual dynamics. OpenAI described Sora as a generalist model of visual data and argued that scaling video generation models may be a promising path toward general-purpose simulators of the physical world [15]. Whether current video models truly understand physics robustly is a separate question. The conceptual direction matters: prediction over video can pressure models to learn temporal, spatial, and physical relations that are not present in static image representation alone [15].

JEPA-like systems point in another direction. I-JEPA predicts representations of target image blocks from context blocks rather than reconstructing pixels, with the goal of learning semantic representations without relying on hand-crafted augmentations [16]. Such systems are not automatically Level 3; many are still best understood as Level 2 representation learners. But they show a path toward predictive representation learning where the target is not raw generation, but future or missing latent structure. If such systems become capable of supporting planning, world modeling, or action-conditioned prediction, they may become part of the broader Level 3 regime [16].

The future of Level 3 may therefore include several branches: LLMs that reason and use tools, multimodal models that connect text with images, audio, video, and interaction, video models that learn physical dynamics, JEPA-like systems that predict latent structure, and embodied systems that learn through closed-loop action and feedback. The common thread is not modality. The common thread is predictive relational generalization [15], [16], [58].

5.9 The Boundary of Level 3

Level 3 is the most powerful level described in this framework, but it has boundaries. It should not be confused with full intelligence, full autonomy, or complete world understanding. Current LLMs can reason, code, explain, summarize, plan, and solve many tasks, but they can also hallucinate, fail under distribution shift, lack persistent agency, misunderstand physical constraints, and depend heavily on prompting, tools, retrieval, memory, or scaffolding. Current video and world-model-like systems can

generate impressive dynamics, but they are still limited in reliable physical understanding and controllable interaction [5], [9]–[12], [15], [16], [58], [60]–[63].

The point of Level 3 is not that the problem of intelligence is solved. The point is that deep learning has entered a new generalization regime. The model is no longer merely learning a task mapping or a transferable representation. It is learning predictive relational structure rich enough that the output stream can become a general task interface. This is a different kind of generalization from Level 1 and Level 2 [4], [5], [10]–[12], [15], [16], [57], [58].

The boundary also explains why not every generative model is automatically Level 3. A small generative model trained on a narrow task may still be Level 1. A masked model that produces representations for downstream adaptation may still be Level 2. A translation model that autoregressively decodes a target sentence may still be Level 1. Level 3 requires breadth, predictive relational structure, and an interface through which learned relations can be used across many task contexts. It is not defined by the mere presence of generation, but by the role prediction plays in organizing and releasing high-dimensional relational structure [2], [4], [5], [19], [22], [42], [57].

The conclusion is therefore clear. Level 1 made deep learning powerful as supervised cross-sample generalization. Level 2 made deep learning foundational as self-supervised representation generalization. Level 3 makes deep learning open-ended by turning predictive modeling into a general interface for behavior, reasoning, tool use, and future world modeling. GPT-style LLMs are the first decisive realization of this level because language and code contain a dense externalization of human cognitive structure. But the level itself is larger than LLMs. It is the general regime of autoregressive and predictive high-dimensional relational generalization [4], [5], [9]–[12], [15], [16], [57], [58], [60], [61].

6. BERT and GPT as Different Generalization Regimes

6.1 Why BERT vs GPT Is Not Merely a Model Comparison

The contrast between BERT and GPT is one of the clearest ways to understand the difference between Level 2 and Level 3. At a superficial level, both belong to the same broad historical period. Both are Transformer-based models. Both use large-scale self-supervised pretraining on text. Both learn from unlabeled corpora rather than relying only on human-labeled downstream datasets. Both helped establish the modern foundation-model era [2]–[5], [49], [57]. If the analysis stops at this level, BERT and GPT may appear to be two variants of the same idea.

But this is not the right level of analysis. BERT and GPT are not merely different architectures. They embody different generalization regimes. BERT represents the mature form of Level 2 in NLP: self-supervised representation generalization. GPT represents the first decisive form of Level 3: autoregressive predictive high-dimensional relational generalization. The deepest difference is not encoder versus decoder as a mechanical architecture label. The deepest difference is how learned structure is exposed and used [2], [4], [5], [19], [57].

BERT learns a representation and then downstream systems use that representation. Its central interface is representation adaptation. GPT learns through prediction and then uses the same prediction process as the task interface. Its central interface is autoregressive continuation. This single difference changes the meaning of pretraining, inference, adaptation, scaling, and capability release [2], [4], [5], [19], [57].

In BERT, the task is usually outside the model’s pretraining interface. A downstream task is attached to the model through a classifier, span predictor, token tagger, reranker, retriever, or fine-tuning setup. The model’s hidden states are the reusable object [2]. In GPT, the task can be written directly into the

model’s context. The model continues the sequence, and the continuation itself can be the answer, explanation, code, reasoning path, translation, summary, plan, or tool call. The model’s output stream is the reusable interface [4], [5], [9], [10], [19], [57], [58].

BERT and GPT therefore should not be collapsed into one undifferentiated category of “large language models.” BERT is large, self-supervised, and foundational, but it is not Level 3 in the central sense of this paper. GPT is also self-supervised, but its autoregressive interface changes what kind of generalization becomes possible. The model no longer only transfers representations across tasks. It can use context itself to specify tasks and release behavior [2], [4], [5], [19], [57].

This section therefore treats BERT and GPT not as two competing model families, but as two different answers to the same historical question: after Level 1 task-specific supervision, what should deep learning generalize next? BERT answered: reusable language representations. GPT answered: predictive relational structure released through generated sequences [2], [4], [5], [57].

6.2 BERT as the Mature Form of Level 2

BERT’s importance lies in making self-supervised representation generalization the default logic of modern NLP. It was designed to pretrain deep bidirectional representations from unlabeled text by conditioning on both left and right context in all layers [2]. Its core training objective, masked language modeling, hides some input tokens and trains the model to recover them using surrounding context. In the original formulation, BERT also used next sentence prediction to train sentence-pair relationships [2].

The key point is that BERT’s prediction objective is not its final user interface. The model predicts masked tokens during pretraining, but after pretraining its main product is the contextual representation. The hidden states of the encoder are then adapted to downstream tasks. A classification task can read the [CLS] representation. A question answering task can use token-level representations to predict answer spans. A sequence labeling task can attach a token classifier. A retrieval or reranking system can use representations to compare texts. The output of BERT’s pretraining process is therefore not a general open-ended text generator. It is a reusable language encoder [2], [19].

This is exactly Level 2. BERT learns structure from data itself, but the learned structure is primarily released through downstream specialization. It does not naturally answer arbitrary questions through the same mechanism by which it was pretrained. It does not normally solve a task by continuing a prompt. It does not make the token stream itself the central task interface. It produces representations that other mechanisms use [2], [19].

BERT’s success made this regime unavoidable. The original paper reported new state-of-the-art results on eleven NLP tasks, including GLUE, MultiNLI, SQuAD v1.1, and SQuAD v2.0 [2]. These results were not important merely because they improved benchmarks. They were important because one pretrained encoder could be adapted across many different task formats. Natural language inference, question answering, sentence-pair classification, and span extraction could all be served by the same pretrained representation base [2], [50], [51].

This changed the workflow of NLP. Before BERT-style pretraining, a downstream task model often carried much of the modeling burden. A researcher or engineer would design the architecture, features, and training setup around the task. After BERT, the pretrained encoder became the main carrier of language structure. Downstream tasks increasingly became specialization layers on top of a shared pretrained base. This is the practical meaning of representation generalization [2], [19], [56].

BERT also shows why Level 2 is more than ordinary supervised learning. The downstream tasks are supervised, but the broad linguistic structure does not come primarily from those downstream labels.

It comes from pretraining on raw text. The downstream task teaches the model how to use the representation for a particular output format, but it does not create the entire linguistic substrate from scratch. The representation already contains syntax, semantics, sentence relations, lexical regularities, and contextual structure learned from unlabeled data [2], [56].

At the same time, BERT’s boundary is clear. It does not become an open-ended reasoning or instruction-following system merely because it is strong. It can support question answering, classification, retrieval, and extraction, but it generally requires task-specific adaptation or a downstream mechanism. Its power is infrastructural. It gives many NLP systems a strong representation base. That is why BERT is a defining landmark of Level 2, not Level 3 [2], [4], [5], [19], [57].

6.3 GPT as the First Decisive Form of Level 3

GPT’s importance lies in making prediction itself the capability interface. GPT-style models are trained autoregressively: given previous tokens, predict the next token. At first, this looks like a simple language modeling objective. But the interface created by this objective is fundamentally different from BERT’s. The model is trained to continue text, and it is also used by asking it to continue text. There is no sharp separation between pretraining behavior and inference behavior [3]–[5], [57].

GPT-2 made the broader implication visible. Its central claim was that language models can function as unsupervised multitask learners [4]. The model was trained on WebText and evaluated on tasks such as question answering, translation, summarization, and reading comprehension in a zero-shot setting [4]. The important point is not only that GPT-2 generated fluent text. The important point is that tasks could begin to be expressed as context. The model did not need a new classifier head for every task. It could read a prompt-like context and produce a continuation [4], [19].

GPT-3 made this regime decisive. GPT-3 was a 175-billion-parameter autoregressive language model evaluated in zero-shot, one-shot, and few-shot settings without task-specific gradient updates or fine-tuning [5]. Tasks and demonstrations were specified through text interaction with the model [5]. This is the central Level 3 breakthrough. The model’s context becomes the task specification. The continuation becomes the task output. The same next-token prediction interface can produce an answer, explanation, classification, translation, summary, analogy, code snippet, or reasoning attempt [5], [9], [19].

This is not just a larger Level 2 model. GPT-3 does not primarily expose its capability through a hidden representation that downstream heads must read. It exposes capability through generation. The prompt tells the model what kind of relation or behavior is required, and the model continues accordingly. A few examples in the context can define a new task format. A natural language instruction can specify a goal. A partial code file can define a programming problem. A dialogue history can define an interaction. A mathematical problem can invite step-by-step reasoning. In each case, the model’s output stream directly participates in the task [5], [9], [10], [19].

GPT belongs to Level 3 because it learns predictive relational structure over broad symbolic data, and that structure is released through autoregressive continuation. The model does not merely generalize representations across tasks. It generalizes task behavior through context. It can use the same interface to adapt to many task formats without changing parameters. This does not make it perfect, reliable, or fully intelligent, but it does make it a different generalization regime [5], [19], [60], [61].

The difference becomes even clearer when code enters the picture. Codex-style models show that autoregressive language modeling over code can produce executable programs from natural language docstrings and context [9]. This is not merely another text classification task. The model generates symbolic structures that can be run, tested, corrected, and connected to external computation. Code

makes visible that GPT-style prediction can release procedural and tool-like structure, not just prose [9].

Instruction tuning then made the GPT interface more usable. InstructGPT showed that larger language models do not automatically follow user intent well, and that supervised demonstrations plus reinforcement learning from human feedback can make models better aligned with instructions [62]. ChatGPT made the interface conversational [63]. Chain-of-thought prompting showed that generated intermediate steps can improve reasoning performance [10]. These developments are natural extensions of the GPT regime because the model’s output stream is already the site where task behavior appears [10], [58], [62], [63].

GPT is therefore not merely “a decoder.” It is a predictive interface model. Its architecture matters, but the deeper significance is the alignment between training and use: train by predicting continuation; use by conditioning on context and generating continuation. This alignment is what allows GPT-style models to become general task interfaces [4], [5], [19], [57].

6.4 The Interface Difference: Representation Adaptation vs Predictive Interaction

The most important difference between BERT and GPT can be summarized as follows: BERT requires representation adaptation; GPT enables predictive interaction [2], [4], [5], [19], [57].

A BERT-style model receives an input and produces contextual hidden states. These hidden states are then used by another mechanism. If the task is sentiment classification, a classifier maps the representation to a label. If the task is extractive question answering, a span head predicts start and end positions. If the task is named entity recognition, a token classifier predicts entity labels. If the task is retrieval, embeddings are compared. The intelligence-like structure learned by BERT is real, but it is accessed through adaptation [2], [19], [50], [51].

A GPT-style model receives an input context and continues it. The task can be expressed inside the context itself. “Translate this sentence into French” is a context. “Answer the following question” is a context. “Here are three examples; infer the pattern” is a context. “Write a Python function that...” is a context. “Explain this theorem step by step” is a context. The model’s continuation is the behavior. No separate classifier head is required for the task to be expressed [4], [5], [9], [10], [19], [57].

This difference changes the role of the user. With BERT, the user or system designer usually decides how to extract and use the representation. With GPT, the user can often specify the task directly in natural language. Prompting therefore becomes central. Prompting is not a superficial trick; it is the control surface of the predictive interface. Surveys of prompt-based learning describe a shift from ordinary supervised input-output learning toward pretrain-prompt-predict paradigms where task specifications are encoded as prompts [19]. That shift is one of the practical signs of Level 3.

The difference also changes what it means to generalize to a new task. In BERT-style Level 2, generalization to a new task often means that the pretrained representation can be fine-tuned or adapted with task data. In GPT-style Level 3, generalization to a new task can mean that the task is described in the prompt and solved by continuation without parameter updates. GPT-3’s few-shot setting made this especially clear: examples are placed in the context, and the model performs the task by continuing the pattern [5], [19].

This does not mean GPT never benefits from fine-tuning or tools. It does. Instruction tuning, RLHF, retrieval, tool use, and agentic scaffolding can all improve GPT-style systems [58], [62]. The point is that GPT’s base interface is already interactive. Fine-tuning improves or aligns the interface; it does not create the interface from scratch [58], [62].

This is the largest conceptual difference between BERT and GPT. BERT makes language structure reusable as representation. GPT makes language prediction usable as interaction. BERT turns text into features. GPT turns text into a task medium [2], [4], [5], [19], [57].

6.5 The Training-Inference Alignment Difference

Another way to understand the BERT-GPT split is through training-inference alignment. In GPT-style models, the training objective and the inference interface are deeply aligned. During training, the model predicts the next token given previous context. During inference, the model predicts the next token given user-provided context. The form is the same. The context changes, but the behavior remains continuation [3]–[5], [57].

This alignment is not trivial. It means that the same operation used to learn from data is also used to act at test time. The model is trained on continuation and used through continuation. When tasks are expressed as text, the training behavior becomes a general task behavior. Autoregressive language modeling could therefore become an open-ended interface [4], [5], [19], [57].

BERT’s training and inference structure is different. During pretraining, BERT predicts masked tokens using bidirectional context. During downstream use, however, the model is often not asked to fill masks. Instead, its hidden states are used by task-specific heads. There is therefore a stronger separation between pretraining objective and downstream capability interface. Masked prediction is a way to shape representations; it is not usually the way the model solves arbitrary tasks at inference time [2], [19].

This does not make BERT weaker in its own regime. For many understanding, retrieval, classification, and extraction tasks, BERT-like encoders are extremely effective. The point is not superiority in all tasks. The point is regime. BERT’s pretraining objective creates reusable hidden states. GPT’s pretraining objective creates a continuation interface. The former is representation-centered; the latter is prediction-centered [2], [4], [5], [56], [57].

This difference also explains why scaling BERT and scaling GPT lead to different historical trajectories. Scaling BERT-like models improves the quality of representations and downstream task performance, but the model still needs adaptation mechanisms. Scaling GPT-like models improves the predictive model, and because prediction is also the interface, capabilities can appear directly in prompting, in-context learning, reasoning traces, code generation, and dialogue. This is one reason GPT-style scaling produced such a dramatic shift in how users interact with models [5], [9], [10], [56], [57], [60], [61].

The difference is not simply that one model is bidirectional and the other is unidirectional. Bidirectionality helps BERT learn strong contextual representations. Autoregression gives GPT a natural generation interface. But the deeper issue is how the model’s learned structure becomes usable. BERT’s structure becomes usable through representations. GPT’s structure becomes usable through continuation [2], [4], [5], [19], [57].

6.6 T5, UL2, and the Attempt to Unify Language Learning

The BERT-GPT distinction is central, but the history is not a simple binary. T5 and UL2 are important because they show that the field recognized the fragmentation among objectives, architectures, and task formats and tried to unify them [64], [65].

T5 introduced a unified text-to-text framework in which every NLP problem is cast as a text-to-text task [64]. Translation, summarization, classification, and question answering can all be represented as input text mapped to output text. This was an important step because it weakened the old separation

between different downstream task formats. Instead of building a separate output head for every task type, T5 framed them all as sequence generation [64].

T5 therefore sits in an interesting position. It is not BERT-like in the narrow encoder-only sense, because it uses an encoder-decoder Transformer and a text-to-text interface. It is also not GPT-like in the pure decoder-only autoregressive sense, because its dominant formulation still involves an encoder-decoder architecture and a transfer-learning setup with pretraining and downstream fine-tuning. T5 helps bridge Level 2 and Level 3, but it does not erase the distinction. It shows that task interfaces can be unified as text, yet the question remains whether the model’s main capability is released through representation adaptation, supervised fine-tuning, or open-ended predictive interaction [19], [64].

UL2 pushed this unification further by proposing a mixture-of-denoisers objective and explicitly disentangling architectural archetypes from pretraining objectives [65]. Its aim was to build models effective across different tasks and setups by combining different denoising paradigms [65]. This matters because it shows that the field was not simply choosing between BERT and GPT as frozen categories. Researchers were actively exploring the space among encoder-only, encoder-decoder, prefix language modeling, denoising, and autoregressive objectives [65], [66].

Yi Tay’s analysis of what happened to BERT and T5 is useful in this context because it directly addresses why BERT-like encoder models did not remain the dominant scaled-up general-purpose route [66]. The argument is not that BERT became useless. Encoder models remain important for retrieval, ranking, classification, embedding, and many understanding tasks. The point is that the field’s center of gravity shifted toward more flexible generative or denoising/autoregressive formats because modern systems increasingly want a general-purpose model interface rather than a separate task-specific model for every setting [66].

This supports the main claim of this section. The historical trajectory after BERT did not simply scale encoder-only representation models into general assistants. Instead, the field moved toward interfaces that could express many tasks as text generation, prompting, continuation, instruction following, or denoising. T5 and UL2 show intermediate and unifying attempts. GPT-style models show the most decisive form of the Level 3 interface [4], [5], [19], [57], [64]–[66].

Therefore, the BERT-GPT distinction should not be interpreted as a rigid taxonomy of architectures. It is a distinction between capability interfaces. BERT-like models organize reusable representations. T5-like models unify many tasks into text-to-text transfer. GPT-like models turn prediction itself into a general interaction surface. These are historically connected, but they are not the same generalization regime [2], [4], [5], [19], [57], [64]–[66].

6.7 Scaling Routes: Why GPT-Like Scaling Changed the Field More Dramatically

Both BERT-like and GPT-like models can scale. But they scale into different kinds of usefulness [5], [56], [57].

Scaling BERT-like encoders improves representation quality. Larger encoders, better pretraining data, better masking strategies, and better optimization can produce stronger performance on GLUE, SQuAD, retrieval, ranking, classification, and extraction tasks. RoBERTa showed that BERT had been undertrained and that better data scale, longer training, dynamic masking, larger batches, and removing next sentence prediction could substantially improve performance [56]. This is real progress. It strengthens Level 2 [2], [56].

But GPT-like scaling changed the field more dramatically because its gains appeared through the same interface users could directly interact with. GPT-2 showed unsupervised multitask behavior

through context [4]. GPT-3 showed few-shot and zero-shot performance without gradient updates [5]. Chain-of-thought prompting showed reasoning improvements through generated intermediate steps [10]. Codex showed that autoregressive models trained on code could generate executable programs from natural language and code context [9]. BIG-bench and emergent-abilities work documented that large models can show new capabilities or sharply improved performance on complex tasks as scale increases [60], [61].

This difference is not merely about benchmark scores. It is about capability access. A stronger BERT encoder may improve many systems, but those systems still need to be built around it. A stronger GPT-like model can expose new capabilities directly through prompting, dialogue, code generation, or reasoning traces. The user can interact with the model as a task medium. GPT-like scaling therefore felt like a broader paradigm shift [5], [9], [10], [19], [60], [61].

The GPT route also changed what “one model” means. In the BERT regime, one model can serve many downstream tasks, but often through different heads, datasets, or fine-tuning procedures. In the GPT regime, one model can receive many task descriptions through the same context window and respond through the same output channel. The same interface supports question answering, summarization, translation, classification, coding, explanation, tutoring, brainstorming, reasoning, and tool-use sequences. This is a much stronger form of interface unification [2], [4], [5], [9], [19], [57].

This does not mean BERT-like models are obsolete or unimportant. For many embedding, retrieval, ranking, classification, and understanding pipelines, encoder-style models remain highly valuable. The point is that they occupy a different layer. They are powerful representation infrastructure. GPT-style models became the center of open-ended interaction because their predictive interface aligns naturally with tasks expressed in language [2], [4], [5], [56], [57], [66].

This is the historical reason GPT became the clearer path to Level 3. It did not merely scale parameters. It scaled a task interface [4], [5], [9], [10], [19], [57], [60], [61].

6.8 Summary of the BERT-like and GPT-like Scaling Routes

Table 2. BERT-like and GPT-like Scaling Routes as Different Generalization Regimes

Core level in this framework

- **BERT-like Route:** Level 2: self-supervised representation generalization
- **GPT-like Route:** Level 3: autoregressive / predictive high-dimensional relational generalization

Main model form

- **BERT-like Route:** Encoder-centered representation model
- **GPT-like Route:** Decoder-centered autoregressive predictive model

Typical pretraining signal

- **BERT-like Route:** Masked token recovery, denoising, representation learning from bidirectional context
- **GPT-like Route:** Next-token prediction over broad text, code, and interaction-like data

Main product of pretraining

- **BERT-like Route:** Contextual hidden representations
- **GPT-like Route:** A predictive continuation interface

Main capability interface

- **BERT-like Route:** Fine-tuning, task heads, span extraction, token labeling, embedding, retrieval, reranking

- **GPT-like Route:** Prompting, in-context learning, generation, dialogue, instruction following, code generation, reasoning traces, tool calls

How a new task is usually expressed

- **BERT-like Route:** Through a downstream dataset, objective, head, or adaptation mechanism
- **GPT-like Route:** Through natural language context, demonstrations, instructions, or interaction history

What scale primarily improves

- **BERT-like Route:** Representation quality and transfer to downstream systems
- **GPT-like Route:** Predictive relational structure and directly accessible task behavior

Typical strengths

- **BERT-like Route:** Understanding, classification, extraction, retrieval, ranking, embedding-based systems
- **GPT-like Route:** Open-ended generation, instruction following, in-context task adaptation, coding, reasoning, tool-use interfaces

Boundary

- **BERT-like Route:** Powerful but usually requires adaptation or a downstream mechanism to produce task behavior
- **GPT-like Route:** Powerful but not automatically reliable, grounded, autonomous, or fully intelligent

Historical meaning

- **BERT-like Route:** Deep learning becomes reusable representation infrastructure
- **GPT-like Route:** Deep learning becomes an open-ended predictive task interface

This table captures the central argument of the section. BERT-like and GPT-like models are not merely two implementation choices inside the same category. They represent different ways of turning self-supervised learning into generalization. BERT-like models generalize by making representations reusable. GPT-like models generalize by making prediction interactive [2], [4], [5], [19], [57], [64]–[66].

The distinction also explains why the history unfolded the way it did. BERT became a foundation for many NLP systems because it supplied strong representations. GPT became the basis of modern assistants because it supplied a direct interface for tasks. T5 and UL2 show that the field explored intermediate forms and unifying objectives, but the deepest contrast remains: representation adaptation versus predictive interaction [2], [4], [5], [57], [64]–[66].

6.9 What the BERT-GPT Split Proves for the Three-Level Framework

The BERT-GPT split proves that the three-level framework is not merely a retrospective classification. It captures a real historical fork in deep learning [2], [4], [5], [57], [64]–[66].

If one only looks at data scale, model scale, or Transformer architecture, BERT and GPT appear similar. Both use self-supervised pretraining. Both benefit from scale. Both learn from text. Both can be adapted to many tasks. But once we ask how the learned structure is used, the difference becomes decisive. BERT’s learned structure is used as representation. GPT’s learned structure is used through prediction. That is a level difference [2], [4], [5], [19], [49], [57].

This is exactly the kind of distinction the three-level framework is designed to reveal. A conventional model-history narrative might say: first came BERT, then came GPT, both are large pretrained language models. That narrative is not false, but it is too shallow. It misses the change in the object

and interface of generalization. BERT generalized representations across tasks. GPT generalized predictive task behavior across contexts [2], [4], [5], [19], [57].

The distinction also prevents two common mistakes. The first mistake is to treat BERT as simply an inferior GPT. That is wrong. BERT is not failed GPT. It is a successful Level 2 model whose purpose is representation. The second mistake is to treat GPT as simply a larger representation learner. That is also wrong. GPT’s central breakthrough is not only that it has hidden representations, but that its autoregressive interface can express and execute tasks [2], [4], [5], [19], [57], [66].

The BERT-GPT comparison is therefore one of the strongest supports for the paper’s thesis. Deep learning progress is not only a matter of bigger models or better benchmarks. It is a matter of what kind of structure the model learns to generalize, and how that structure becomes usable. BERT and GPT learned from similar broad raw material, but they turned that material into different generalization regimes [2], [4], [5], [19], [57], [64]–[66].

The conclusion is direct. BERT is the defining NLP landmark of Level 2 because it made reusable language representation the center of modern NLP. GPT is the defining landmark of Level 3 because it made predictive continuation a general task interface. This difference is not cosmetic. It is one of the clearest historical demonstrations that deep learning advances by changing the object of generalization [2], [4], [5], [19], [57].

7. What the Three-Level Framework Explains

7.1 Deep Learning Progress as a Sequence of Generalization Transitions

The central claim of this paper is that deep learning should be understood as a machinery of generalization. Its major historical transitions are not merely transitions in architecture, benchmark performance, dataset size, or parameter count. They are transitions in what kind of structure the model learns to generalize.

Level 1 established supervised cross-sample generalization. In this regime, deep models learn task-specific mappings from labeled data and generalize those mappings to unseen samples. AlexNet made this level historically decisive in computer vision by showing that a large supervised CNN could dominate ImageNet-scale natural image classification and defeat feature-engineered visual recognition systems [1]. Early neural NLP extended the same level into language and sequence tasks, showing that neural networks could learn distributed linguistic structures and supervised sequence mappings, including neural machine translation [38], [39], [42], [43].

Level 2 established self-supervised representation generalization. In this regime, the central product of learning is no longer a task-specific mapping, but a reusable representation. BERT made this shift clear in NLP by pretraining deep bidirectional language representations from unlabeled text and adapting them across many downstream tasks [2]. SimCLR, DINO, MAE, and CLIP extended the same logic into vision and multimodality: models could learn transferable visual or image-text representations from data-derived supervision rather than relying only on fixed category labels [21]–[23], [45], [54], [55].

Level 3 established autoregressive and predictive high-dimensional relational generalization. In this regime, the model’s predictive process becomes a general task interface. GPT-style language models made this transition decisive by learning from broad text and code distributions and releasing learned relational structure through continuation, prompting, instruction following, reasoning traces, code generation, and tool-use sequences [4], [5], [9]–[12], [57], [58], [62], [63]. In Level 3, prediction is not merely a narrow task output or a representation-learning pressure. It becomes the medium through which learned structure is used.

This framework explains why the history of deep learning cannot be reduced to “models got bigger.” Models did get bigger, and scale was often essential [57], [67]. But scale mattered differently at each level. In Level 1, scale made supervised task mappings powerful enough to dominate high-dimensional perception. In Level 2, scale made representations transferable enough to become reusable infrastructure. In Level 3, scale made predictive models broad enough that the prediction interface could support open-ended task behavior.

The same surface variable—scale—therefore has different meanings depending on the generalization regime. A larger CNN classifier, a larger BERT encoder, and a larger GPT-style autoregressive model are not simply three points on one flat curve. They scale different objects: a task mapping, a representation, and a predictive interface. The three-level framework is therefore necessary.

7.2 Why Architecture Alone Is Not the Story

The framework also explains why architecture-only narratives are insufficient. Architecture matters. CNNs mattered for vision. Transformers mattered for language. Self-attention made many later models more scalable and flexible [49]. Residual learning, attention, encoder-decoder modeling, masking, contrastive objectives, and autoregression all changed what models could learn and how efficiently they could learn it. But architecture alone does not explain the historical transitions.

AlexNet was not historically decisive only because it was a CNN. CNNs existed before AlexNet [25]. Its significance came from the alignment of architecture, scale, GPU computation, data, training recipe, and a visible benchmark where the result broke the previous empirical order [1]. BERT was not historically decisive only because it used the Transformer. Its significance came from turning a Transformer encoder into a general pretraining-and-fine-tuning representation substrate for NLP [2]. GPT was not historically decisive only because it used a Transformer decoder. Its significance came from aligning autoregressive training and inference into a predictive interface that could absorb task specification into context [4], [5], [57].

This is the deeper lesson: architectures become historically transformative when they enable a new generalization regime. A convolutional architecture helped Level 1 vision because it provided the right inductive bias for images. A Transformer encoder helped Level 2 NLP because it made deep contextual representation learning scalable. A Transformer decoder helped Level 3 because autoregressive continuation became a natural interface for language, code, instruction, reasoning, and interaction.

This also prevents a common mistake in interpreting deep learning history. It is easy to look backward and say that progress came from “better architectures.” That is partly true, but incomplete. The more important question is: what did the architecture make generalizable? Did it make a supervised mapping scale? Did it make a reusable representation possible? Did it make predictive interaction possible? The framework focuses on this deeper object.

7.3 Why Scale Matters, but Only Through the Right Regime

The framework also clarifies the role of scale. Sutton’s “bitter lesson” argues that the most effective long-term AI methods are those that leverage computation and search/learning, rather than relying primarily on human-designed knowledge [68]. Deep learning history strongly reflects this lesson. AlexNet, BERT, GPT-3, CLIP, and modern reasoning models all depended on learning from data at scale rather than encoding most structure manually [1], [2], [5], [10]–[12], [23].

But this paper’s framework adds an important refinement: scale is not a single generic force. Scale reveals different things depending on what the model is trained to generalize. In Level 1, scale reveals the strength of learned task-specific mappings. In Level 2, scale reveals the transfer power of learned

representations. In Level 3, scale reveals the breadth of predictive relational structure that can be released through an output interface.

Scale alone is therefore not enough. Chinchilla showed that model size and training data must be balanced under a compute budget, and that some earlier large language models were undertrained relative to the amount of data used [67]. This supports a more precise view of scale: it is not just parameter count. It is the alignment of model size, data scale, compute, objective, optimization, and interface.

AlexNet was not merely large; it was trainably large under the right visual task and compute regime. BERT was not merely large; it was a large pretrained encoder whose representations could transfer across tasks. GPT-3 was not merely large; it was a large autoregressive model whose predictive interface could support in-context task behavior [1], [2], [5]. The meaning of scale is regime-dependent.

Future progress should therefore not be expected from scale in the abstract. It will depend on what is being scaled. Scaling a weak objective may produce limited gains. Scaling a narrow interface may produce narrow usefulness. Scaling a broad predictive interface over rich data can unlock new behavior. Scaling multimodal or world-predictive systems may unlock different kinds of generalization. The decisive question is not only how much scale is used, but what generalization object is being scaled.

7.4 What the Framework Adds to Existing Narratives

Existing narratives often describe deep learning history through architectures: CNNs, RNNs, LSTMs, Transformers, diffusion models, large language models. Others describe it through datasets and benchmarks: MNIST, ImageNet, GLUE, SQuAD, MMLU, BIG-bench. Others emphasize hardware and scaling: GPUs, distributed training, model size, data scale, compute laws. These narratives are all useful, but they remain partial.

The three-level framework reorganizes these facts around a deeper axis: what is generalized?

This axis explains why AlexNet, BERT, and GPT are not merely three successful models. They are regime markers. AlexNet marks the public dominance of large-scale supervised cross-sample generalization in high-dimensional perception. BERT marks the dominance of self-supervised representation transfer in language. GPT marks the emergence of predictive interaction as a general task interface. Each model matters not only because it performed well, but because it made a different kind of generalization historically visible.

The framework also explains why later models often look like natural continuations. VGG, GoogLeNet, and ResNet refined Level 1 supervised vision. RoBERTa, SimCLR, DINO, MAE, and CLIP refined or expanded Level 2 representation learning. InstructGPT, ChatGPT, chain-of-thought prompting, Toolformer, o1-style reasoning models, and DeepSeek-R1-style reinforcement learning for reasoning all refine or expand Level 3 predictive interaction [10]–[12], [21]–[23], [35]–[37], [45], [54]–[56], [58], [62], [63].

In this view, deep learning history becomes less chaotic. It is not a random sequence of models. It is a sequence of increasing generalization scope: from samples, to representations, to predictive relational behavior.

8. Implications for the Further Development of Deep Learning Generalization

8.1 Future Progress Will Come from New Generalization Objects

If this framework is correct, future progress in deep learning will not come only from larger versions of current models. Larger models will matter, but the most important advances will come from discovering new generalization objects and training regimes.

Level 1 generalized task-specific mappings. Level 2 generalized reusable representations. Level 3 generalized predictive relational structures through open-ended interfaces. Future progress can continue inside each level. Level 1 will remain important wherever tasks are clearly defined, labels are reliable, and outputs must be controlled. Level 2 will continue to improve representation infrastructure for retrieval, ranking, perception, multimodal alignment, scientific data, robotics perception, and domain-specific systems. Level 3 will continue to expand predictive interfaces for reasoning, coding, tool use, dialogue, planning, and richer forms of interaction.

The deeper point is that the next major advances may come from extending the object of generalization beyond the current frontier. GPT-style LLMs are extremely powerful because language and code externalize a large amount of human cognitive structure. But they do not automatically solve grounding, persistent agency, long-horizon planning, embodied interaction, or reliable world modeling [9], [13], [14], [17], [69], [70]. These problems may require richer predictive regimes, richer memory systems, richer interaction loops, and tighter coupling between perception, action, and feedback.

The future of deep learning may therefore be less about replacing the three levels and more about developing them further. Supervised task generalization, representation generalization, and predictive relational generalization will likely remain useful layers in larger systems. A mature AI system may combine all three: supervised heads for controlled tasks, representation models for retrieval and perception, and predictive models for open-ended interaction. The most important question remains what structure is being generalized and through what interface.

8.2 World Models as an Early but Important Post-LLM Level 3 Frontier

World models should be understood as one of the most important current frontiers after the success of LLMs. This does not mean that the route is already clear. It is not. The term “world model” is used across several research traditions, including model-based reinforcement learning, video generation, predictive representation learning, robotics, and embodied AI [13]–[17], [69], [70]. The field is still early, fragmented, and conceptually unsettled. But the reason it matters is clear: if Level 3 is predictive relational generalization, then learning predictive structure over world states, actions, dynamics, and consequences is a natural next domain where the Level 3 idea becomes important.

Current world-model-related work already appears in several concrete forms. In model-based reinforcement learning, DreamerV3 learns a model of the environment and improves behavior by imagining future scenarios; it reports strong performance across a broad range of control tasks using a single configuration [14]. This line shows one concrete sense of “world model”: a learned predictive model that supports behavior and control.

In video generation, the world-model idea appears differently. OpenAI described Sora as a generalist model of visual data and argued that scaling video generation models is a promising path toward general-purpose simulators of the physical world [15]. This does not prove that current video models have reliable physical understanding. It means that video prediction and generation are being explicitly connected to the problem of learning temporal, spatial, object-level, and physical

regularities from visual data. In the language of this paper, video generation is interesting because it may pressure models to learn predictive relations that static image representation alone does not require.

In predictive representation learning, JEPA-like systems provide another existing form. I-JEPA predicts representations of target image blocks from context blocks rather than reconstructing raw pixels, aiming to learn semantic image representations without relying on hand-crafted augmentations [16]. V-JEPA extends feature prediction to video, training models using a feature prediction objective over video and evaluating representations on image and video tasks [71]. These systems are not automatically Level 3 in their current form. Many are still best understood as Level 2 or transitional predictive representation systems. But they show that world-related prediction does not have to mean pixel-level reconstruction; it can also mean predicting latent structure.

The current state is therefore important but early. World models are not yet a settled route beyond LLMs. There is no single agreed architecture, training objective, data regime, evaluation standard, or capability interface. Model-based RL, video generation, JEPA-style predictive representation, and embodied systems all touch the world-model problem from different angles. Their relationship to one another is still being worked out.

The correct claim should therefore be strong but careful. World models are not a solved successor to LLMs. They are not yet a mature Level 3 regime comparable to GPT-style language models. But they are one of the most important areas where the next development of Level 3 may be searched for, because they move predictive generalization from text and code toward dynamics, action, perception, and consequences.

8.3 Modality as the Further Development of Level 2 and Level 3

Modality should be understood as an axis along which Level 2 and Level 3 continue to develop. It is not a separate level by itself. A model does not become more advanced merely because it has more modalities. The key question remains the same: what is generalized, and how is the learned structure used?

In Level 2, modality expansion mainly appears as representation alignment and representation transfer. CLIP is the cleanest example. It learns image and text encoders aligned in a shared representation space, allowing visual concepts to be specified through natural language rather than only through fixed supervised labels [23]. This is a major Level 2 development because the learned object is not a single classifier but a transferable image-text representation substrate.

The same Level 2 logic extends into stronger visual and multimodal representation systems. DINO and MAE showed that self-supervised vision can produce transferable visual representations through self-distillation and masked reconstruction [22], [55]. DINOv2 further pushed the ambition of producing general-purpose visual features without supervision [72]. These systems show that modality expansion is not just about adding images to language. It also includes building stronger representations inside each modality and across modalities.

In Level 3, modality expansion has a different meaning. The goal is no longer only to align representations, but to extend the predictive interface. A multimodal Level 3 system can receive images, video, audio, or sensor data as part of the context and generate text, actions, explanations, code, or other outputs through an interactive interface. Flamingo showed that visual-language models can handle interleaved image/video and text inputs and perform few-shot learning on open-ended visual-language tasks through prompting [73]. GPT-4V showed that users could instruct GPT-4 to analyze image inputs [74]. GPT-4o was described as an autoregressive omni model that accepts combinations of text, audio, image, and video and generates combinations of text, audio, and image outputs [75]. These systems show modality becoming part of the predictive interface.

Embodied and robotic models push modality expansion further. PaLM-E incorporates continuous sensor modalities into a language model and applies the resulting embodied multimodal model to embodied reasoning, visual-language tasks, and language tasks [69]. RT-2 incorporates web-scale vision-language models into robotic control by expressing actions as tokens and co-fine-tuning on robotic trajectory data and vision-language tasks [70]. These systems do not prove that multimodal models have solved embodiment, but they show a concrete direction: perception and action can be brought into the same representational or predictive interface.

This gives modality a precise role in the three-level framework. In Level 2, modality expansion means learning representations that align or transfer across sensory and symbolic domains. In Level 3, modality expansion means extending the predictive interface so that tasks can involve images, video, audio, sensors, actions, and interaction histories. The same word “multimodal” can therefore refer to different generalization regimes. A CLIP-like model is multimodal Level 2 because its central achievement is representation alignment. A GPT-4V-like or GPT-4o-like system is closer to multimodal Level 3 because the modality enters an interactive predictive interface.

This distinction matters because multimodality can easily be misunderstood. More modalities do not automatically mean more intelligence. A multimodal classifier can still be narrow. A multimodal representation model can still be Level 2. A multimodal agent can still fail if it lacks reliable prediction, grounding, control, or feedback [17], [69], [70]. The significance of modality depends on whether it expands the object of generalization and whether it strengthens the interface through which learned structure becomes usable.

9. Scope, Boundaries, and Misreadings

9.1 This Framework Is Not a Linear Ranking of Model Worth

The levels should not be misread as a simple ranking where Level 3 makes Level 2 obsolete and Level 2 makes Level 1 obsolete. The levels describe different generalization regimes, not a hierarchy in which later models automatically replace earlier ones.

Level 1 remains essential. Supervised models are still necessary for many domains where tasks are clear, labels are reliable, and evaluation is specific. Level 2 remains essential. Representation models are still central for retrieval, ranking, embedding, classification, segmentation, detection, and many production systems. Level 3 is powerful, but it is not always the right tool. Open-ended generation may be unnecessary, inefficient, or unsafe for many tasks.

The point is not that later levels replace earlier levels. The point is that later levels expand the scope of what can be generalized. A mature AI system may combine all three: supervised task heads, reusable representations, and predictive interactive models. The levels are often layered in real systems.

9.2 This Framework Does Not Deny the Importance of Architecture, Optimization, or Engineering

Another possible misreading is that if generalization is the central story, architecture and engineering do not matter. This is false. Architecture, optimization, hardware, data pipelines, distributed systems, evaluation, safety, alignment, and product design all matter enormously.

The claim is not that architecture is irrelevant. The claim is that architecture matters because it enables a generalization regime. CNNs mattered because they made visual structure learnable at scale. Transformers mattered because they made sequence modeling and contextual representation learning scalable [49]. Autoregressive decoders mattered because they created a predictive continuation

interface. Engineering mattered because these regimes would not work without stable training, efficient computation, data scale, and system integration.

This is also why the “bitter lesson” should not be interpreted as “ideas do not matter” [68]. The correct lesson is more subtle. Human-designed knowledge should not be overbuilt into brittle systems when scalable learning can discover structure from data. But the choice of objective, architecture, data, interface, and training regime is still a deep scientific and engineering problem. The best ideas are often those that make scalable learning possible.

9.3 This Framework Is Historical, but Also Predictive

The framework is built from history, but it is not only retrospective. It also suggests what to look for next. A future breakthrough will likely not be identifiable only by a new architecture name or a larger benchmark score. It will be identifiable by a new object of generalization or a new capability interface.

A future model may deserve a new level if it learns a structure that current levels do not capture: persistent world-state generalization, action-grounded predictive planning, autonomous memory-based learning, or some other form of system-level generalization. But until such a regime becomes clear, Level 3 describes the current frontier: predictive relational generalization released through open-ended interfaces.

This makes the framework useful as a research lens. It asks researchers to look beyond surface metrics and ask: what is being generalized? Across what boundary? Through what interface? With what training signal? Under what scale regime? These questions are more durable than any single benchmark.

10. Conclusion

Deep learning is fundamentally a machinery of generalization.

Its history is not best understood as a loose sequence of architectures or benchmarks. It is better understood as a sequence of transitions in what models can generalize. The first transition made deep learning powerful as supervised cross-sample generalization. The second made deep learning foundational as self-supervised representation generalization. The third made deep learning open-ended as autoregressive and predictive high-dimensional relational generalization.

Level 1 established that deep neural networks could learn task-specific structures from labeled data and generalize them to unseen samples. In vision, AlexNet made this undeniable by breaking the scale barrier of supervised CNNs on ImageNet [1]. In language, early neural NLP and neural machine translation showed that neural networks could learn distributed linguistic structure and supervised sequence mappings [38], [39], [41]–[43].

Level 2 established that deep learning could move beyond task-specific systems and become reusable representation infrastructure. BERT showed that self-supervised pretraining could produce language representations transferable across many tasks [2]. SimCLR, DINO, MAE, and CLIP showed that vision and multimodality could also be reorganized around transferable representations and alignment [21]–[23], [45], [54], [55].

Level 3 established that predictive modeling itself could become a general task interface. GPT-style models showed that next-token prediction over broad language and code distributions could produce systems capable of in-context learning, instruction following, reasoning traces, code generation, dialogue, and tool-use behavior [4], [5], [9]–[12], [57], [58], [62], [63]. This does not mean that LLMs are full human minds or complete AGI. It means they are the first decisive realization of a new deep learning generalization regime.

The BERT-GPT split makes the framework especially clear. BERT and GPT both come from the era of large-scale self-supervised Transformer pretraining, but they do not represent the same level. BERT makes representations reusable. GPT makes prediction interactive. BERT is the defining NLP landmark of Level 2. GPT is the defining landmark of Level 3.

The broader lesson is that deep learning advances when models learn to generalize a new kind of structure. The field should therefore not ask only whether a model is larger, faster, or better on a benchmark. It should ask what the model has learned to generalize, how that structure is exposed, and what new interface for capability has been created.

The next era of deep learning will likely be shaped by this same principle. Progress may come from richer world models, broader modality integration, stronger representation infrastructure, more reliable predictive interfaces, action-conditioned learning, or new forms of system-level generalization. But whatever the concrete path, the deepest question will remain the same:

What is being generalized?

That question is the organizing thread of deep learning's past, and it is likely to remain the organizing thread of its future.

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